

The Three-Dimensional Navier–Stokes Regularity Problem

Thomas Birnie

August 23, 2026

Abstract

We prove global smoothness for the unforced three-dimensional incompressible Navier–Stokes equations on $\mathbb{R}^3$ for every viscosity $\nu > 0$ and every divergence-free Schwartz initial velocity u_0 . The corresponding velocity and pressure remain smooth for all $t \geq 0$.

The proof rules out a finite singular endpoint directly. Let $[0, T_*)$ be the maximal classical lifespan and suppose $T_* < \infty$. Repeated time differentiation of the Navier–Stokes equations, with pressure recovered at each order by Riesz transforms, generates one pressure-complete mixed jet of the same solution. For each fixed finite temporal depth N and spatial Sobolev height M , the central estimate controls the corresponding rectangle of that jet uniformly over every cutoff $T < T_*$. The adjacent-row trace estimate then gives, for every finite r and m ,

$$\sup_{0 \leq t < T_*} \|\partial_t^r u(t)\|_{H^m(\mathbb{R}^3)} < \infty.$$

These finite rectangles agree under restriction. Their compatible intersection preserves the whole-terminal control at every finite order, and the same trace mechanism gives a strong terminal limit in every Sobolev space:

$$u(t) \longrightarrow u_* \quad \text{in } H^m(\mathbb{R}^3) \quad \text{as } t \uparrow T_*$$

for every finite m , with one terminal velocity

$$u_* \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3).$$

The differentiated Navier–Stokes equations and the pressure recovery determine the complete velocity-pressure jet at T_* . Thus the alleged singular endpoint is a smooth time slice of the same solution. Smooth local existence restarts the system from u_* , and uniqueness identifies that restart with the original solution. The same solution then continues beyond T_* , contradicting maximality. Hence $T_* = \infty$.

Contents

1	The Regularity Problem	1
2	Critical Height and the Derivative Tower	2
2.1	Vortex Stretching	2
2.2	The temporal Tower	5
2.3	More spatial regularity, not less	6
3	The Intersection Theorem	10
3.1	Why the finite pieces are rectangles	10
3.2	Every finite rectangle remains bounded	14
4	From the Intersection to Global Smoothness	16
4.1	The first coordinate already closes the endpoint	16
4.2	The critical-height roof	17
4.3	The two vortex readings	19
4.4	No finite singular time	21
5	Further Consequences and the Euler Boundary	22
5.1	A finite sum across infinitely many scales	22
5.2	Three classical criteria inside one low rectangle	23
5.3	The local singularity picture closes as well	24
5.4	Why the Euler Tower is different	24
A	Local Construction and Classical Estimates	26
A.1	The local construction behind the maximal history	26
A.2	The Classical Estimates Before Closure	30
B	Derivative and Finite-Record Calculus	35
B.1	Critical Height at a Finite Endpoint	35
B.2	What finite time implies after one response escapes	36
B.3	The Pressure-Complete Mixed Jet	37
B.4	The Spatial Column Inside Critical Height	38
B.5	Simultaneous Sobolev Intersections	40
B.6	Finite Adjacent Rectangles	42
C	Compatible Terminal Assembly	45
C.1	The Terminal Quantifier	45
C.2	The Compatible Intersection	48
C.3	The Critical Height at the Terminal Face	49
C.4	One Endpoint for the Whole Tower	51
C.5	Restarting the Same History	52
	Bibliography	54

1 The Regularity Problem

The three-dimensional Navier–Stokes regularity problem asks whether a smooth solution can lose its smoothness in finite time. The velocity u and pressure p must satisfy

$$\begin{aligned} \partial_t u + (u \cdot \nabla)u + \nabla p &= \nu \Delta u, & \nabla \cdot u &= 0 & \text{on } \mathbb{R}^3 \times (0, \infty), \\ u(\cdot, 0) &= u_0 & \text{on } \mathbb{R}^3. \end{aligned} \quad (1.1)$$

Here u is the $\mathbb{R}^3$ -valued velocity, p is the scalar pressure, and $\nu > 0$ is the fixed kinematic viscosity.

Fefferman’s whole-space formulation assumes that the initial velocity u_0 in (1.1) is smooth, divergence-free, and rapidly decreasing together with all of its spatial derivatives (Fefferman 2000). In standard notation, this means

$$\begin{aligned} \mathcal{S}_\sigma &:= \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3) \cap \ker(\nabla \cdot), & u_0 &\in \mathcal{S}_\sigma, \\ \forall \alpha \in \mathbb{N}_0^3 \, \forall K \in \mathbb{N}_0 \, \exists C_{\alpha,K} < \infty \, \forall x \in \mathbb{R}^3, & |\partial_x^\alpha u_0(x)| &\leq C_{\alpha,K} (1 + |x|)^{-K}. \end{aligned} \quad (1.2)$$

Here $\mathcal{S}(\mathbb{R}^3; \mathbb{R}^3)$ is the Schwartz space of smooth $\mathbb{R}^3$ -valued functions. The symbol α is a spatial multi-index, and K is the prescribed algebraic decay order. For the fixed datum u_0 , the constant $C_{\alpha,K}$ may depend on u_0 , α , and K , but not on x .

For finite $m \in \mathbb{N}_0$, write

$$H_\sigma^m(\mathbb{R}^3) := H^m(\mathbb{R}^3; \mathbb{R}^3) \cap \ker(\nabla \cdot).$$

Fix $K > 3/2$. For every $|\alpha| \leq m$, (1.2) gives

$$\|\partial_x^\alpha u_0\|_2^2 \leq C_{\alpha,K}^2 \int_{\mathbb{R}^3} (1 + |x|)^{-2K} dx < \infty.$$

There are only finitely many multi-indices with $|\alpha| \leq m$, so $u_0 \in H_\sigma^m(\mathbb{R}^3)$ for every finite m . Taking $m = 0$ gives $u_0 \in L^2(\mathbb{R}^3)$, so the initial kinetic energy is finite.

Fix this u_0 and a viscosity $\nu > 0$. Let (u, p) be the corresponding classical solution on its maximal smooth interval $[0, T_*)$, where $T_* \in (0, \infty]$.

The whole-space claim is that this same solution remains smooth for all time and has uniformly finite kinetic energy:

$$u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)), \quad \sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty. \quad (1.3)$$

The energy clause is already controlled throughout the maximal lifespan: Proposition A.2.1 gives $\|u(t)\|_2 \leq \|u_0\|_2$ for every $0 \leq t < T_*$. The regularity problem is to rule out a finite endpoint of the smooth history:

$$T_* = \infty.$$

2 Critical Height and the Derivative Tower

The momentum equation has four pieces:

$$\underbrace{\partial_t u}_{\text{local acceleration}} + \underbrace{(u \cdot \nabla)u}_{\text{nonlinear transport}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}.$$

The central regularity problem lies in the competition between nonlinear transport and viscosity. The hope is simple: viscosity smooths the fluid faster than nonlinear transport can sharpen it. If that remains true at every scale, the velocity stays smooth.

2.1 Vortex Stretching

The surrounding flow pulls a narrow vortex outward along its axis. On a short material segment with spatially uniform strain, let $e(t)$ be its unit tangent in a positive strain direction:

$$S := \frac{1}{2}(\nabla u + (\nabla u)^\top), \quad Se(t) = \sigma(t)e(t), \quad \sigma(t) > 0.$$

The material-line law gives its length $L(t)$ by

$$\frac{\dot{L}(t)}{L(t)} = e(t) \cdot Se(t) = \sigma(t).$$

The vortex lengthens. But it is the same incompressible fluid. Its material volume $\mathcal{V}$ cannot change. With $A(t) := \mathcal{V}/L(t)$ and $r_{\text{eff}}(t) := \sqrt{A(t)/\pi}$, incompressibility gives

$$\nabla \cdot u = 0, \quad \frac{d}{dt}(AL) = 0, \quad \frac{\dot{A}}{A} = -\sigma(t), \quad \frac{\dot{r}_{\text{eff}}}{r_{\text{eff}}} = -\frac{\sigma(t)}{2}.$$

As the length grows, the effective cross-section falls and the vortex becomes narrower.

The same axial strain acts on the rotation carried by the vortex. With $\omega := \nabla \times u$,

$$\frac{D\omega}{Dt} = S\omega + \nu \Delta \omega, \quad \frac{D}{Dt} := \partial_t + u \cdot \nabla.$$

Along the tracked trajectory, the alignment $\omega = |\omega|e(t)$ gives

$$S\omega = \sigma(t)\omega, \quad \frac{1}{2} \frac{D|\omega|^2}{Dt} = \sigma(t)|\omega|^2 + \nu \omega \cdot \Delta \omega.$$

Stretching contributes $\sigma(t)|\omega|^2 > 0$. Viscosity has no fixed pointwise sign; to stop the amplification, it must offset that contribution. When their sum remains positive along the material trajectory, $|\omega|$ grows while r_{eff} falls.

$$\frac{|\omega|}{\sqrt{2}} = \left| \frac{1}{2}(\nabla u - (\nabla u)^\top) \right|_{\text{F}}.$$

The rotational gradients rise.

Navier–Stokes Scaling

The radius is falling while stretching and viscosity are still competing over the same rotation. At a time $t_0 < T_*$, write $\ell := r_{\text{eff}}(t_0)$. For $\lambda > 1$, the exact Navier–Stokes comparison at width $\lambda^{-1}\ell$ is

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t).$$

At time $\lambda^{-2}t_0$, the corresponding element in u_λ has effective radius $\lambda^{-1}\ell$. This is another solution at another scale, not a later state of the original vortex. Each momentum term transforms with it:

$$\begin{aligned} \partial_t u_\lambda(x, t) &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), \\ (u_\lambda \cdot \nabla) u_\lambda(x, t) &= \lambda^3 ((u \cdot \nabla) u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda(x, t) &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), \\ \nu \Delta u_\lambda(x, t) &= \lambda^3 \nu (\Delta u)(\lambda x, \lambda^2 t). \end{aligned}$$

Every term carries the same λ^3 , so the finer width gives viscosity no relative advantage over nonlinear sharpening.

Critical Height

With $\Lambda := (-\Delta)^{1/2}$, the same rescaling gives

$$\widehat{u}_\lambda(\xi, t) = \lambda^{-2} \widehat{u}(\lambda^{-1} \xi, \lambda^2 t),$$

and a change of variables gives

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2.$$

At $s = 0$, concentration lowers kinetic energy. At $s = 1$, it raises the first-derivative norm. Between them, the exponent vanishes at $s = \frac{1}{2}$. Set

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2.$$

Writing H_λ for the same quantity evaluated on u_λ ,

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \|u(\lambda^2 t)\|_2^2, \quad H_\lambda(t) = H(\lambda^2 t), \quad \|\nabla u_\lambda(t)\|_2^2 = \lambda \|\nabla u(\lambda^2 t)\|_2^2.$$

The comparison can move the structure to a finer width, but it cannot move H .

Nonlinear Production versus Viscous Dissipation

If H rises, the original equation has done it. Its signed nonlinear contribution is

$$\mathcal{P}_{1/2}[u](t) := -\operatorname{Re} \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} ((u(t) \cdot \nabla) u(t)) \right\rangle_{L^2}.$$

The pressure pairing vanishes because $\nabla \cdot u = 0$, and the Laplacian contributes $-\nu \|\Lambda^{3/2}u\|_2^2$, leaving

$$H'(t) + \nu \|\Lambda^{3/2}u(t)\|_2^2 = \mathcal{P}_{1/2}[u](t).$$

The height rises only when nonlinear production exceeds viscous dissipation. Under the same rescaling,

$$\mathcal{P}_{1/2}[u_\lambda](t) = \lambda^2 \mathcal{P}_{1/2}[u](\lambda^2 t), \quad \nu \|\Lambda^{3/2}u_\lambda(t)\|_2^2 = \lambda^2 \nu \|\Lambda^{3/2}u(\lambda^2 t)\|_2^2.$$

Production and dissipation again gain the same power. Their common λ^2 has moved the unresolved contest onto the shortened time coordinate $t \mapsto \lambda^{-2}t$.

The Heat Clock

Viscosity carries its own version of that contracted clock. For the linear heat equation $v_t = \nu \Delta v$,

$$\hat{v}(\xi, t + \delta t) = e^{-\nu|\xi|^2 \delta t} \hat{v}(\xi, t).$$

A scale-localized structure of width r occupies frequencies $|\xi| \simeq r^{-1}$. Its viscous change becomes order one when

$$\nu|\xi|^2 \delta t \simeq 1,$$

so the linear comparison time is

$$\tau_\nu(r) := \frac{r^2}{\nu}.$$

At width $\lambda^{-1}r$,

$$\tau_\nu(\lambda^{-1}r) = \lambda^{-2} \tau_\nu(r).$$

The finer comparison has not strengthened viscosity relative to sharpening. It has shortened the clock for both by the same square.

The Zeno Cascade

Repeated halving makes those clocks a geometric sequence. For

$$r_n := 2^{-n} r_0, \quad \tau_n := \frac{r_n^2}{\nu},$$

the comparison clocks satisfy

$$\tau_n = \frac{r_0^2}{\nu} 4^{-n}, \quad \sum_{n=0}^{\infty} \tau_n = \frac{4r_0^2}{3\nu} < \infty.$$

The clocks are summable. That arithmetic leaves room for infinitely many finer comparisons before a finite time, but it does not turn them into one fluid history. For these clocks to

describe a candidate singularity, one Navier–Stokes solution would have to pass through widths

$$r_0 > r_1 > r_2 > \cdots \longrightarrow 0$$

at times satisfying the additional schedule

$$t_0 < t_1 < t_2 < \cdots \uparrow T_* < \infty, \quad t_{n+1} - t_n \asymp \tau_n,$$

with comparison constants independent of n . Under that hypothesis, the remaining time is

$$T_* - t_n \asymp \sum_{k=n}^{\infty} \tau_k = \frac{4r_n^2}{3\nu}.$$

At the n -th width, both the next transition and the total time left are of order r_n^2/ν . The same velocity field would have to change across intervals that collapse to zero with its spatial scale.

2.2 The temporal Tower

The velocity may be growing or sharpening, and the rate at which it changes has a rate of change of its own. The same reading continues through every later response. At a fixed spatial coordinate, the sequence is

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \partial_t^3 u, \quad \dots$$

Each row is the Eulerian time derivative of the one before it. They are successive responses of the same field, not separate singularity scenarios. Together they form the temporal Tower. Write

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad n \in \mathbb{N}_0. \quad (2.2.1)$$

Every finite differentiation below is taken on a strict interval where the solution remains classical.

Navier–Stokes tells us exactly what those rows contain. For $V_0 = u$,

$$V_1 = \nu \Delta V_0 - (V_0 \cdot \nabla) V_0 - \nabla Q_0.$$

One time derivative can land on either copy of velocity in the transport term:

$$V_2 = \nu \Delta V_1 - (V_1 \cdot \nabla) V_0 - (V_0 \cdot \nabla) V_1 - \nabla Q_1.$$

One more derivative produces the middle placement twice:

$$V_3 = \nu \Delta V_2 - (V_2 \cdot \nabla) V_0 - 2(V_1 \cdot \nabla) V_1 - (V_0 \cdot \nabla) V_2 - \nabla Q_2.$$

At order n , the binomial coefficients count every way the time derivatives can divide between the transporting velocity and the velocity being transported:

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n, \quad \nabla \cdot V_n = 0. \quad (2.2.2)$$

This recurrence is the exact Eulerian temporal Tower generated by Navier–Stokes.

Pressure appears in every row as well. Taking the divergence of the original equation and using decay at spatial infinity recovers it from the velocity:

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j). \quad (2.2.3)$$

Here $R_j := \partial_j (-\Delta)^{-1/2}$, and the decaying normalization removes the arbitrary spatial constant in pressure at each time. After n time derivatives,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.2.4)$$

The velocity rows determine Q_n , and ∇Q_n appears in the equation for V_{n+1} . The temporal Tower is already spatial inside the equation. Viscosity contributes ΔV_n , transport contributes a spatial derivative, and pressure is recovered through spatial singular integrals of the same velocity products. The resulting mixed coordinates are

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p, \quad \alpha \in \mathbb{N}_0^3. \quad (2.2.5)$$

One spatial derivative splits the transport term again:

$$\partial_\ell \left((V_a \cdot \nabla) V_{n-a} \right) = ((\partial_\ell V_a) \cdot \nabla) V_{n-a} + (V_a \cdot \nabla) \partial_\ell V_{n-a}.$$

For a general multi-index α , Leibniz's rule gives

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} \\ = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned} \quad (2.2.6)$$

The pressure coordinates satisfy

$$-\Delta \Pi_{n,\alpha} = \partial_x^\alpha \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.2.7)$$

Following the candidate singularity through successive responses led down the time direction, and the equation has opened spatial derivatives inside every rung. Pressure stays tied to the same rows throughout. The next count is the spatial height carried by each rung.

2.3 More spatial regularity, not less

The temporal Tower follows the candidate singularity through one response after another. Inside the critical-height calculation, every differentiation in time opens another step up in space. Viscosity inserts a Laplacian, and the scale weight of $H^{(n)}$ matches the Sobolev energy at height $n + \frac{1}{2}$. The Tower is asking for more spatial regularity, not less. It has only exposed that regularity so far; it has not bounded it. All identities below are taken on an interval where the solution is classical, so the differentiations and pairings are justified.

For

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad E_{s,\lambda}(t) := \frac{1}{2} \|\Lambda^s u_\lambda(t)\|_2^2,$$

the appendix scaling calculation in Proposition B.1.1 gives

$$E_{s,\lambda}(t) = \lambda^{2s-1} E_s(\lambda^2 t). \quad (2.3.1)$$

Since $H = E_{1/2}$,

$$H_\lambda(t) = H(\lambda^2 t).$$

After n time derivatives,

$$H_\lambda^{(n)}(t) = \lambda^{2n} H^{(n)}(\lambda^2 t). \quad (2.3.2)$$

The spatial energy at height $n + \frac{1}{2}$ carries exactly the same factor:

$$E_{n+1/2,\lambda}(t) = \lambda^{2n} E_{n+1/2}(\lambda^2 t). \quad (2.3.3)$$

Equal scale weight does not identify the two quantities. It places the spatial energy $E_{n+1/2}$ beside the critical response $H^{(n)}$.

Applying Λ^s to Navier–Stokes and pairing the result with $\Lambda^s u$ gives the viscous contribution

$$\nu \langle \Lambda^s u, \Lambda^s \Delta u \rangle_{L^2} = -\nu \left\| \Lambda^{s+1} u \right\|_2^2 = -2\nu E_{s+1}.$$

Since the global pressure pairing vanishes against the divergence-free velocity, define the nonlinear production at height s by

$$\mathcal{P}_s := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u \right) \right\rangle_{L^2}.$$

The complete spatial-energy row is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}. \quad (2.3.4)$$

The first few derivatives let us watch the spatial ladder appear. At critical height, one time derivative reaches the next spatial energy:

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

Differentiate again and use the same row at $s = \frac{3}{2}$:

$$\begin{aligned} H'' &= \partial_t \mathcal{P}_{1/2} - 2\nu E'_{3/2} \\ &= \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

The third derivative reaches one height farther:

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

Each differentiation keeps every transfer term already produced and exposes one higher viscous energy. For $n \geq 1$, induction gives

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}. \quad (2.3.5)$$

Because $\nu > 0$, the top spatial energy can be isolated:

$$E_{n+1/2} = (-2\nu)^{-n} \left(H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2} \right). \quad (2.3.6)$$

This is the exact gain and the exact limitation. The induction proved in Proposition B.4.1 isolates the higher Sobolev energy, but does not bound it. A bound requires control of the complete right-hand side of (2.3.6).

The temporal Tower enters through $H^{(n)}$ itself. Since

$$H(t) = \frac{1}{2} \left\langle \Lambda^{1/2} V_0(t), \Lambda^{1/2} V_0(t) \right\rangle_{L^2},$$

Leibniz's rule gives

$$H^{(n)}(t) = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{n-a}(t) \right\rangle_{L^2}. \quad (2.3.7)$$

The three entries on one rung remain different objects. The field response is V_n , the scalar response $H^{(n)}$ contracts pairs of temporal rows, and the viscous calculation exposes the spatial energy $E_{n+1/2}$.

Table 2.3.1: The first time–space rungs of the critical-height calculation.

order	Eulerian response	critical response	spatial height
0	$V_0 = u$	$H = E_{1/2}$	$E_{1/2}$
1	$V_1 = u_t$	H'	$E_{3/2}$
2	$V_2 = u_{tt}$	H''	$E_{5/2}$
3	$V_3 = u_{ttt}$	H'''	$E_{7/2}$
n	$V_n = \partial_t^n u$	$H^{(n)}$	$E_{n+1/2}$

The temporal column records the successive responses of the candidate singularity. The spatial column records the higher Sobolev energies exposed beside them: $E_{1/2}, E_{3/2}, E_{5/2}, E_{7/2}, \dots$

This is one exposed ladder, not yet a regularity estimate. The transfer terms can still carry the size of its highest rung, and the calculation has so far opened the ladder only inside the original velocity response $V_0 = u$.

The same calculation can be made inside every temporal response. For $r \in \mathbb{N}_0$, let $E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2$, and define the complete transport–pressure transfer in the r -th recurrence by

$$\mathcal{P}_{r,s} := -\operatorname{Re} \left\langle \Lambda^s V_r, \Lambda^s \left[\sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right] \right\rangle_{L^2}.$$

Pairing the r -th temporal equation with $\Lambda^{2s} V_r$ gives

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \quad (2.3.8)$$

After any finite ascent depth $k \geq 1$,

$$\partial_t^k E_{r,1/2} = (-2\nu)^k E_{r,k+1/2} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{r,j+1/2}. \quad (2.3.9)$$

The index r chooses how far the singularity has been followed in time, while k chooses how high the spatial ladder has been followed inside that response. These are independent coordinates of the same mixed jet.

One spatial ladder inside every temporal response forms the grid

	H_x^0	H_x^1	H_x^2	H_x^3	$\dots$
$V_0 = u$	•	•	•	•	$\dots$
$V_1 = u_t$	•	•	•	•	$\dots$
$V_2 = u_{tt}$	•	•	•	•	$\dots$
$V_3 = u_{ttt}$	•	•	•	•	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

The bullets mark derivatives of the same velocity field, not bounds already proved for them. Moving down chooses a later Eulerian response, and moving across chooses a higher Sobolev height inside that response. Fix $T < T_*$ and set $I = (0, T)$. Control of every coordinate on this strict classical interval would place the velocity in the mixed intersection

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)). \quad (2.3.10)$$

The grid is infinite, but any chosen temporal order and spatial height lie inside a finite block. Fix N and M ; the rows through N and the columns through M form one rectangle. For each fixed rectangle, the datum-generated whole-terminal estimate proved in Proposition C.1.2 controls the entire rectangle uniformly over $T < T_*$; its bound may depend on N and M . Every coordinate appears in some rectangle, and the rectangles agree wherever they overlap. That agreement allows Section 3 to pass from the finite rectangles to their compatible intersection.

3 The Intersection Theorem

3.1 Why the finite pieces are rectangles

Section 2 ended with a finite block of the mixed grid. Let (u, p) be the maximal classical solution generated by the fixed pair (u_0, ν) , and write its lifespan as $[0, T_*)$. On the finite-lifespan branch used below, set

$$I_* := (0, T_*).$$

To turn its spatial coordinates into time traces, pair each response V_n with its time derivative V_{n+1} . Let $A = (1 - \Delta)^{1/2}$. For a smooth Hilbert-valued function v and $0 \leq a < b \leq T$,

$$\|v(b)\|_{H^M}^2 - \|v(a)\|_{H^M}^2 = 2 \operatorname{Re} \int_a^b \langle A^{M-1} v_t(t), A^{M+1} v(t) \rangle_{L^2} dt. \quad (3.1.1)$$

Cauchy–Schwarz now places the change of the middle norm between its two adjacent space-time rows:

$$\left| \|v(b)\|_{H^M}^2 - \|v(a)\|_{H^M}^2 \right| \leq 2 \|v_t\|_{L^2(a,b;H^{M-1})} \|v\|_{L^2(a,b;H^{M+1})}.$$

Consequently,

$$v \in L^2(0, T; H^{M+1}), \quad v_t \in L^2(0, T; H^{M-1}) \quad (3.1.2)$$

gives

$$v \in C([0, T]; H^M). \quad (3.1.3)$$

Time mollification and Fourier truncation justify (3.1.1) for the pair in (3.1.2); weak H^M continuity together with continuity of the norm gives the strong representative.

For $M = 2$, the pair is already concrete:

$$u \in L^2(0, T; H^3), \quad u_t \in L^2(0, T; H^1) \implies u \in C([0, T]; H^2). \quad (3.1.4)$$

The upper row reaches one derivative above the desired middle space. The next temporal row reaches one derivative below it. Their pairing carries u through the middle.

Apply this to $v = V_n$, with $v_t = V_{n+1}$. If we want the first $N + 1$ temporal responses to arrive in H^M , the squared velocity norm used to measure that finite block is

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) &:= \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned} \quad (3.1.5)$$

For $M = 0$, the lower edge is read in H^{-1} . Temporal depth N and spatial height M remain independent.

For example, $(N, M) = (1, 2)$ contains two overlapping trace pairs:

Table 3.1.1: The $(1, 2)$ finite rectangle. The row V_1 is shared: it is the time derivative used to trace V_0 , and it is also the next response whose own trace is paired with V_2 .

temporal row	one step above	one step below
$n = 0$	$V_0 = u \in L^2 H^3$	$V_1 = u_t \in L^2 H^1$
$n = 1$	$V_1 = u_t \in L^2 H^3$	$V_2 = u_{tt} \in L^2 H^1$

Write $\mathcal{R}_{N,M}[u_0; T]$ for this finite pressure-complete block of the mixed jet: the velocity coordinates measured by (3.1.5), together with the corresponding Q_n -coordinates reconstructed by (2.2.7). Every entry is a coordinate of the datum-generated velocity and pressure fields.

From the mixed-jet matrix to the Sobolev rectangles

We can now say exactly what the matrix form records. The mixed recurrence in (2.2.6) and the Sobolev grid in Section 2.3 are two readings of the same derivative family, but they do different jobs. The first records how one differentiated transport row is assembled from the mixed jet. Let

$$\Gamma := \mathbb{N}_0 \times \mathbb{N}_0^3, \quad \kappa = (n, \alpha), \quad \lambda = (a, \beta), \quad \binom{\kappa}{\lambda} := \binom{n}{a} \binom{\alpha}{\beta},$$

and write

$$\lambda \preceq \kappa \iff a \leq n \text{ and } \beta \leq \alpha, \quad \kappa - \lambda = (n - a, \alpha - \beta).$$

Set $J_\kappa := J_{n,\alpha}$ and $\mathbf{J} := (J_\lambda)_{\lambda \in \Gamma}$. The transport blocks are

$$\mathbb{L}(J)_{\kappa,\lambda} := \binom{\kappa}{\lambda} \mathbb{P}[(J_{\kappa-\lambda} \cdot \nabla) \cdot], \quad \lambda \preceq \kappa,$$

with zero blocks when $\lambda \not\preceq \kappa$. Their action is not left implicit. The κ -th block row gives

$$\begin{aligned} [\mathbb{L}(J)\mathbf{J}]_\kappa &= \sum_{\lambda \preceq \kappa} \binom{\kappa}{\lambda} \mathbb{P}[(J_{\kappa-\lambda} \cdot \nabla) J_\lambda] \\ &= \sum_{\lambda \preceq \kappa} \binom{\kappa}{\lambda} \mathbb{P}[(J_\lambda \cdot \nabla) J_{\kappa-\lambda}]. \end{aligned}$$

The second line is the change of index $\lambda \leftrightarrow \kappa - \lambda$, using $\binom{\kappa}{\kappa-\lambda} = \binom{\kappa}{\lambda}$. It is exactly the Leray projection of the nonlinear sum in (2.2.6), so the projected velocity row is

$$J_{n+1,\alpha} + [\mathbb{L}(J)\mathbf{J}]_\kappa = \nu \Delta J_{n,\alpha}.$$

The projection removes $\nabla \Pi_{n,\alpha}$ from this velocity row because $\mathbb{P} \nabla \Pi_{n,\alpha} = 0$; the pressure coordinate remains reconstructed by (2.2.7) and retained in the pressure-complete rectangle.

Order Γ by the total mixed depth $n + |\alpha|$, with any fixed rule for ties. The condition $\lambda \preceq \kappa$ then places the supported blocks on or below the diagonal. This triangularity belongs to the transport bookkeeping: the endpoint terms $\lambda = 0$ and $\lambda = \kappa$ still contain J_κ , while the full equation also contains the forward temporal shift and the Laplacian. Each block is a first-order transport operator followed by the order-zero Leray projection; it depends on the solution and is generally nonlocal.

The Pascal weights count how the derivatives divide between the two velocity factors. They disappear if each coordinate is measured with its own factorial. With $\alpha! := \alpha_1! \alpha_2! \alpha_3!$, set

$$\hat{J}_{n,\alpha} := \frac{J_{n,\alpha}}{n! \alpha!}, \quad \hat{\Pi}_{n,\alpha} := \frac{\Pi_{n,\alpha}}{n! \alpha!}.$$

The cancellation and the surviving temporal factor can be seen in one summand:

$$\begin{aligned} \frac{J_{n+1,\alpha}}{n! \alpha!} &= (n+1) \hat{J}_{n+1,\alpha}, \\ \frac{1}{n! \alpha!} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} &= (\hat{J}_{a,\beta} \cdot \nabla) \hat{J}_{n-a,\alpha-\beta}. \end{aligned}$$

Indeed, $\binom{n}{a} a! (n-a)! = n!$, and the same componentwise identity holds for $\binom{\alpha}{\beta} \beta! (\alpha-\beta)! = \alpha!$. Dividing (2.2.6) by $n! \alpha!$ now gives

$$(n+1) \hat{J}_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} (\hat{J}_{a,\beta} \cdot \nabla) \hat{J}_{n-a,\alpha-\beta} + \nabla \hat{\Pi}_{n,\alpha} = \nu \Delta \hat{J}_{n,\alpha}.$$

Let D be the diagonal factorial matrix $D_{\kappa,\kappa} := \kappa! := n! \alpha!$. Row by row—equivalently, on every finite truncation—the same cancellation reads

$$D^{-1} \mathbb{L}(D \hat{J}) D = \hat{\mathbb{L}}(\hat{J}), \quad \hat{\mathbb{L}}(\hat{J})_{\kappa,\lambda} := \mathbb{P}[(\hat{J}_{\kappa-\lambda} \cdot \nabla) \cdot], \quad \lambda \preceq \kappa.$$

Each normalized block now depends on the difference $\kappa - \lambda$, and every block row contains finitely many terms. This is the lower-triangular Toeplitz, or convolution, form of the transport array. The factor $n+1$ remains on the temporal shift, and pressure remains attached through $\hat{\Pi}_{n,\alpha}$; the normalization has reorganized the original equation, not replaced it with a linear one.

The Sobolev grid gives the second matrix reading by measuring this same jet rather than listing each spatial multi-index separately. For $m \in \mathbb{Z}$, write

$$G_{n,m}(T) := \|V_n\|_{L^2(0,T;H^m)} = \left(\int_0^T \|A^m V_n(t)\|_2^2 dt \right)^{1/2}.$$

Since $J_{n,\alpha} = \partial_x^\alpha V_n$, the integer Sobolev norm satisfies

$$G_{n,m}(T)^2 \asymp_m \sum_{|\alpha| \leq m} \|J_{n,\alpha}\|_{L^2((0,T) \times \mathbb{R}^3)}^2, \quad m \in \mathbb{N}_0.$$

Thus α labels one spatial derivative in the mixed-jet array, whereas m labels the normed envelope that collects the spatial derivatives through order m . With this notation, (3.1.5) is exactly

$$\mathfrak{R}_{N,M}(u; T) = \sum_{n=0}^N G_{n,M+1}(T)^2 + \sum_{n=0}^N G_{n+1,M-1}(T)^2.$$

The rectangle is a finite norm window of the jet; it is not a principal submatrix of $\mathbb{L}(J)$, nor does it add every entry beneath one corner. For each $0 \leq n \leq N$, it selects the staggered trace stencil

$$(n, M+1), \quad (n+1, M-1) \quad \longrightarrow \quad (n, M).$$

The two input coordinates are the adjacent rows in (3.1.5), and the output coordinate is the continuous middle trace produced by (3.1.1). One step down the temporal direction spans two spatial indices and lands between them. Increasing N stacks more of these stencils; changing M moves the same trace mechanism across the spatial columns. The critical-height calculation in (2.3.6) follows one spatial ladder inside the original response $V_0 = u$. Equation (2.3.9) repeats that calculation inside each V_r , opening the corresponding ladder in every temporal row, while the rectangles keep temporal depth and spatial height independent.

The operator array records how the mixed derivatives interact; the Sobolev grid records the sizes of those same derivatives in finite norm windows. Both constructions come from the original pressure-complete history, but they are not the same matrix under a relabelling. When a larger window contains a smaller one, a retained coordinate is the same distributional derivative of the same velocity or normalized pressure, viewed through the lower Sobolev height. That identity, rather than the Toeplitz pattern by itself, is what makes restriction of the finite windows exact. The rectangle theorem is a statement about these normed derivative coordinates, not about the operator norm, spectrum, or invertibility of $\mathbb{L}(J)$. The matrix form turns the Leibniz bookkeeping into a lower-triangular convolution with finitely many terms in each mixed-index row and identifies exactly which finite coordinates enter each trace. It proves no estimate for any coordinate and improves no Sobolev norm. The whole-terminal bound is the separate statement realized in Proposition C.1.2; no sum over all derivative orders and no constant shared by every rectangle is needed.

For any strict cutoff $T < T_*$, classical smoothness makes every fixed block finite:

$$\mathfrak{R}_{N,M}(u; T) < \infty \quad \text{for finite } N, M. \quad (3.1.6)$$

Classicality supplies only the statement on the left:

$$\forall T < T_* : \quad \mathfrak{R}_{N,M}(u; T) < \infty \quad \not\Rightarrow \quad \sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty.$$

The next subsection states the right-hand requirement as Theorem 3.2.1: after N and M have been fixed, one bound must survive every cutoff.

3.2 Every finite rectangle remains bounded

Theorem 3.2.1 (Datum-generated whole-terminal rectangles). *Let $u_0 \in \mathcal{S}_\sigma$, let $\nu > 0$, and let (u, p) be the maximal classical solution generated by (u_0, ν) on $[0, T_*)$. If $T_* < \infty$, then for every finite $N, M \in \mathbb{N}_0$,*

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \quad (3.2.1)$$

The associated pressure rows are generated by (2.2.4) and use the fixed decaying normalization.

Remark 3.2.2 (Reading the terminal quantifier). The pair (N, M) is fixed before the supremum over $T < T_*$ is taken. The bound may depend on (N, M) ; no constant uniform over all derivative orders is asserted. The theorem controls every finite rectangle of the same history, and those rectangles agree under restriction.

The mixed jet has already fixed $\mathcal{R}_{N,M}[u_0; T]$ for each finite pair and cutoff $T < T_*$. The theorem keeps its squared velocity norm bounded independently of that cutoff.

Fix N and M . Every term in (3.1.5) is the integral over $(0, T)$ of a nonnegative squared norm. For any one of those integrands f , the intervals $(0, T)$ exhaust I_* and

$$\int_{I_*} f(t) dt = \lim_{T \uparrow T_*} \int_0^T f(t) dt.$$

The whole-terminal bound makes this limit finite for every row in the fixed rectangle, so

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(I_*; H^{M-1}), \quad 0 \leq n \leq N. \quad (3.2.2)$$

The calculation in Proposition C.1.2 applies the pressure recurrence, Riesz-transform bounds, and finite-order Sobolev products to give the corresponding finite pressure coordinates

$$Q_n \in L^1(I_*; H^M), \quad 0 \leq n \leq N. \quad (3.2.3)$$

These whole-interval velocity and pressure coordinates form $\mathcal{R}_{N,M}[u_0]$ on I_* .

Let the ambient finite-window space be

$$\mathcal{X}_{N,M} := \prod_{n=0}^N L^2(I_*; H^{M+1}) \times \prod_{n=0}^N L^2(I_*; H^{M-1}) \times \prod_{n=0}^N L^1(I_*; H^M),$$

whose three factors carry $(V_n)_{n=0}^N$, $(V_{n+1})_{n=0}^N$, and $(Q_n)_{n=0}^N$, respectively. Thus $\mathcal{R}_{N,M}[u_0] \in \mathcal{X}_{N,M}$.

Choose $N' \leq N$ and $M' \leq M$. The restriction $\rho_{N',M'}^{N,M} : \mathcal{X}_{N,M} \rightarrow \mathcal{X}_{N',M'}$ discards the temporal rows above N' and uses the canonical Sobolev embeddings

$$H^{M+1} \hookrightarrow H^{M'+1}, \quad H^{M-1} \hookrightarrow H^{M'-1}, \quad H^M \hookrightarrow H^{M'}$$

on the retained velocity and pressure coordinates. The underlying distributions are unchanged, so

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (3.2.4)$$

This ordinary overlap is what lets all finite blocks be read at once. In the matrix language above, their compatible intersection is the projective limit of the finite windows:

$$\mathcal{I}[u_0] := \left(\mathcal{R}_{N,M}[u_0] \right)_{N,M \in \mathbb{N}_0} \in \varprojlim_{N,M} \mathcal{X}_{N,M}, \quad \rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (3.2.5)$$

The limit symbol records compatibility under these restriction maps, not convergence in one norm over all derivative orders. There is no new field at the intersection. Given any finite temporal order and any finite spatial height, choose one rectangle large enough to contain them. A still larger rectangle returns the same coordinate of the same u .

Proposition 3.2.3 (The mixed intersection). *The velocity component of $\mathcal{I}[u_0]$ satisfies*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (3.2.6)$$

In particular,

$$u \in \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (3.2.7)$$

Proof. Fix finite r, m , and choose $N \geq r$ and $M \geq m$. The upper edges of that rectangle give

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

These are the distributional time derivatives of the same velocity field, so $u \in H^r(I_*; H^m)$. Since r and m were arbitrary, all finite coordinates belong to the intersection. Taking $r = 0$ gives (3.2.7). $\square$

4 From the Intersection to Global Smoothness

Only a small part of the intersection is needed to contradict a finite endpoint. The lowest adjacent pair reaches the critical norm that would have to blow up. The remaining coordinates explain what becomes of the critical height, the shrinking vortex, the derivative Tower, and the terminal history once that contradiction is closed.

4.1 The first coordinate already closes the endpoint

Start with the velocity row that opened the singularity argument. Choose $(N, M) = (0, 1)$ in (3.2.2). Then

$$u \in L^2(I_*; H^2), \quad u_t \in L^2(I_*; L^2).$$

The adjacent trace identity carries these two rows into the missing middle space:

$$u \in C([0, T_*]; H^1).$$

Since $H^1(\mathbb{R}^3) \hookrightarrow L^3(\mathbb{R}^3)$,

$$\sup_{0 \leq t \leq T_*} \|u(t)\|_3 < \infty. \quad (4.1.1)$$

This is where the finite maximal-time criterion is met by the first rectangle. The endpoint theorem of Escauriaza, Seregin, and Šverák says that a uniform L^3 bound continues the solution through T_* (Escauriaza, G. A. Seregin, and Šverák 2003). Under the standing contradiction hypothesis,

$$T_* < \infty \implies \sup_{0 < t < T_*} \|u(t)\|_3 = \infty. \quad (4.1.2)$$

Equation (4.1.1) gives the opposite conclusion. The lowest critical rectangle already puts the singularity and the estimate on the same velocity, in the same norm, over the same terminal interval.

The base row has already closed the finite endpoint. The remaining coordinates show what the intersection carries with it once that endpoint is closed. Fix $r, m \in \mathbb{N}_0$, choose $N \geq r$, and take the rectangle of middle height $M = m$. Its adjacent rows give

$$V_r \in L^2(I_*; H^{m+1}), \quad V_{r+1} = \partial_t V_r \in L^2(I_*; H^{m-1}).$$

Theorem C.3.1 gives

$$\sup_{0 \leq t \leq T_*} \|\partial_t^r u(t)\|_{H^m} < \infty. \quad (4.1.3)$$

Nothing in this step couples r to m . We may choose any finite temporal response and then ask for any finite spatial height inside that same row.

Given $k \in \mathbb{N}_0$ and $2 \leq p \leq \infty$, choose an integer m with

$$m > k + \frac{3}{2} - \frac{3}{p}. \quad (4.1.4)$$

The spatial Sobolev embedding (Cerdà 2010, Theorem 7.28) gives

$$\left\| \nabla^k \partial_t^r u(t) \right\|_{L^p} \leq C_{k,m,p} \left\| \partial_t^r u(t) \right\|_{H^m}. \quad (4.1.5)$$

Combining this with (4.1.3),

$$\nabla^k \partial_t^r u \in L^\infty(I_*; L^p(\mathbb{R}^3)). \quad (4.1.6)$$

Since T_* is finite under the standing contradiction hypothesis,

$$\nabla^k \partial_t^r u \in L^q(I_*; L^p(\mathbb{R}^3)) \quad (1 \leq q \leq \infty). \quad (4.1.7)$$

The pressure is carried by these same finite coordinates. For fixed $r, m \in \mathbb{N}_0$, the normalized pressure formula (2.2.4) contains a finite sum of products of the rows $V_0, \dots, V_r$. The adjacent traces place those rows continuously in every finite Sobolev space, so Sobolev multiplication and the Riesz-transform bounds give

$$\partial_t^r p = Q_r \in C([0, T_*]; H^m(\mathbb{R}^3)). \quad (4.1.8)$$

For finite k , $2 \leq p \leq \infty$, and $1 \leq q \leq \infty$, choose $m > k + \frac{3}{2} - \frac{3}{p}$. The same spatial embedding (Cerdà 2010, Theorem 7.28) and the finiteness of T_* then give

$$\nabla^k \partial_t^r p \in L^q(I_*; L^p(\mathbb{R}^3)). \quad (4.1.9)$$

Each requested norm uses one finite rectangle, and the compatible intersection keeps these finite coordinates together without replacing them by an all-orders sum. The contradiction already occurs in the base row: the continuation criterion requires $V_0 = u$ to escape in L^3 , while the coordinate $(0, 1)$ places that same velocity continuously in $H^1 \hookrightarrow L^3$. The temporal and spatial directions have met inside the row that carries the original singularity question, and they give opposite conclusions.

4.2 The critical-height roof

The same H^1 trace does more than block the L^3 escape. Since $H = \frac{1}{2} \left\| \Lambda^{1/2} u \right\|_2^2$, it puts a finite roof over the entire terminal height:

$$H_{\max} := \sup_{0 \leq t \leq T_*} H(t) \leq \frac{1}{2} \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^1}^2 < \infty. \quad (4.2.1)$$

Fix $H_0 > 0$, set $L_k := 2^k H_0$ for $k \in \mathbb{N}_0$, and let

$$N_{\text{levels}} := \#\left\{k \in \mathbb{N}_0 : \text{there is a } t \in [0, T_*] \text{ with } H(t) = L_k\right\}.$$

Only finitely many of these dyadic levels can be reached beneath the roof, and

$$N_{\text{levels}} \leq \begin{cases} 0, & H_{\max} < H_0, \\ 1 + \lfloor \log_2(H_{\max}/H_0) \rfloor, & H_{\max} \geq H_0. \end{cases} \quad (4.2.2)$$

No estimate on the duration of an individual crossing is needed. The critical-height ladder is finite, which rules out the unbounded height sequence forced by a finite maximal time. This alone does not rule out shrinking spatial widths; the two vortex readings below test those separate geometric claims.

The next low rectangle, $\mathcal{R}_{0,2}$, controls how the height moves beneath that roof:

$$u \in L^2(I_*; H^3), \quad u_t \in L^2(I_*; H^1), \quad u \in C([0, T_*]; H^2). \quad (4.2.3)$$

Indeed,

$$H'(t) = \operatorname{Re} \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle_{L^2},$$

and Cauchy–Schwarz in time gives

$$\int_0^{T_*} |H'(t)| dt \leq \left\| \Lambda^{1/2} u \right\|_{L_t^2 L_x^2} \left\| \Lambda^{1/2} u_t \right\|_{L_t^2 L_x^2} < \infty. \quad (4.2.4)$$

Also $E_{3/2} \in L^1(I_*)$, so the first completed row

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2}$$

shows that the signed critical production is integrable on I_* . Integrating that same row identifies the signed prefix exactly:

$$\int_0^t \mathcal{P}_{1/2}(s) ds = H(t) - H(0) + 2\nu \int_0^t E_{3/2}(s) ds. \quad (4.2.5)$$

The critical-height roof and the finite $E_{3/2}$ integral give

$$A_*(T_*) := \sup_{0 < t < T_*} \int_0^t \mathcal{P}_{1/2}(s) ds < \infty. \quad (4.2.6)$$

The scalar signed-prefix bound is now a consequence of the compatible intersection. It was not used to produce the mixed jet, its finite rectangles, or their whole-terminal bounds. The height can rise and fall sharply, but it cannot accumulate an infinite net rise before T_* .

The same rectangle also gives the physical control needed for the vortex. In three dimensions, $H^3 \hookrightarrow W^{1,\infty}$, and hence

$$\int_0^{T_*} \|\nabla u(t)\|_\infty dt \leq T_*^{1/2} \|u\|_{L^2(I_*; H^3)} < \infty.$$

Since $\omega = \nabla \times u$ and $S = \frac{1}{2}(\nabla u + \nabla u^\top)$, we obtain the full finite stretching action

$$\int_0^{T_*} (\|\nabla u(t)\|_\infty + \|\omega(t)\|_\infty + \|S(t)\|_\infty) dt < \infty. \quad (4.2.7)$$

4.3 The two vortex readings

The mixed intersection lets us test two distinct readings of a narrowing core. They are not an exhaustive classification of singular geometry. They isolate two different claims a vortex picture can make: either the visible core follows fixed material labels, or its membership changes as the concentration moves through the fluid.

Begin with fixed material labels. Let $X(a, t)$ solve

$$\partial_t X(a, t) = u(X(a, t), t), \quad X(a, 0) = a.$$

For two trajectories in that core and $t < T_*$,

$$\frac{d}{dt} |X(a, t) - X(b, t)| \geq -\|\nabla u(t)\|_\infty |X(a, t) - X(b, t)|.$$

The finite action in (4.2.7) gives

$$|X(a, t) - X(b, t)| \geq |a - b| \exp\left(-\int_0^t \|\nabla u(s)\|_\infty ds\right) > 0 \quad (4.3.1)$$

The same $L^2(I_*; H^3)$ row gives $u \in L^1(I_*; L^\infty)$, so each trajectory extends continuously to T_* . Passing to the limit above gives

$$|X(a, T_*) - X(b, T_*)| \geq |a - b| \exp\left(-\int_0^{T_*} \|\nabla u(s)\|_\infty ds\right) > 0.$$

A material core begins with distinct particles, and those particles remain distinct at the terminal time. The drawn radius may shrink, but it cannot represent the collapse of that same labelled core to one point.

Keep this first reading and now put the vortex exactly on its centreline. The finite coordinate $(r, m) = (0, 3)$ in (4.1.3) gives

$$L_* := \sup_{0 \leq t \leq T_*} \|\nabla u(t)\|_\infty < \infty. \quad (4.3.2)$$

At a chosen centre $x_c(t)$,

$$|u(t, x) - u(t, x_c)| \leq L_* |x - x_c|.$$

Let $C_r(t)$ be a circle of radius r in a cross-section normal to the vortex axis. The constant vector $u(t, x_c)$ integrates to zero around the closed curve, so

$$\left| \oint_{C_r(t)} u(t, x) \cdot d\ell \right| \leq 2\pi L_* r^2 \longrightarrow 0 \quad (r \downarrow 0). \quad (4.3.3)$$

Perfect centring does not evade the Lipschitz bound. If the radius tends to zero while the velocity remains Lipschitz, the relative tangential speed is $O(r)$ and the circulation is $O(r^2)$; no nonzero circulation survives at the limiting point.

The second reading lets the visible core recruit new material as earlier labels leave it. Particle separation no longer follows the changing membership of that core, so the test moves to the whole-field high-frequency tail. For $s > 1/2$ and $K \geq 1$, define

$$H_{>K}(t) := \frac{1}{2} \int_{|\xi| > K} |\xi| |\widehat{u}(t, \xi)|^2 d\xi.$$

Then

$$H_{>K}(t) \leq \frac{1}{2} K^{1-2s} \|u(t)\|_{\dot{H}^s}^2. \quad (4.3.4)$$

Choose an integer $m \geq s$. Equation (4.1.3) with $r = 0$, together with $H^m \hookrightarrow \dot{H}^s$, gives

$$\lim_{K \rightarrow \infty} \sup_{0 \leq t \leq T_*} H_{>K}(t) = 0. \quad (4.3.5)$$

Even when material is continually recruited, the complete velocity field cannot retain a fixed amount of critical content beyond every frequency.

These two readings now have separate conclusions. Fixed material labels retain positive separation at T_* , and a centred radius carries vanishing circulation as it shrinks. Changing membership cannot preserve a nonzero whole-field critical tail beyond every frequency.

4.4 No finite singular time

Theorem 4.4.1 (Whole-space existence and smoothness). *For every $\nu > 0$ and every $u_0 \in \mathcal{S}_\sigma$, the maximal classical solution of (1.1) satisfies*

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)). \quad (4.4.1)$$

For every $t \geq 0$,

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (4.4.2)$$

Proof. Assume that $T_* < \infty$. Theorem 3.2.1 bounds every finite pressure-complete rectangle of the mixed jet uniformly through I_* . The restriction identity (3.2.4) keeps their shared coordinates identical, so Theorem C.2.1 gives

$$u \in \bigcap_{r, m < \infty} H^r(I_*; H^m(\mathbb{R}^3)).$$

Fix m . Taking $r = 1$ places u in $H^1(I_*; H^m)$, and Lemma C.4.1 gives a limit $u_*^{(m)} \in H^m$ as $t \uparrow T_*$. If $k > m$, the limits found in H^k and H^m agree after the embedding $H^k \hookrightarrow H^m$, since both are limits of the same $u(t)$. The compatible limits define one divergence-free endpoint

$$u_* := \lim_{t \uparrow T_*} u(t) \in H_\sigma^\infty(\mathbb{R}^3) \subset \bigcap_{m < \infty} H^m(\mathbb{R}^3). \quad (4.4.3)$$

This is the common endpoint constructed in Lemma C.4.1.

The same argument reaches every temporal row. For finite N, m , take $r = N + 1$ in the intersection. Then $V_N = \partial_t^N u \in H^1(I_*; H^m)$, so it has a terminal trace $V_N^* \in H^m$; compatibility in m places V_N^* in H^∞ . Passing to the limit in the pressure and velocity recurrences gives

$$\begin{aligned} Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \\ V_{N+1}^* &= \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \end{aligned}$$

Thus the left-hand history reaches T_* with the actual normalized velocity and pressure jet generated from u_* , as proved in Proposition C.4.2.

Apply Lemma C.5.1 at T_* with datum u_* . It gives a smooth solution on $[T_*, T_* + \delta]$. Its right-hand jet is generated from the same u_* by the same normalized recurrences, so every derivative agrees at the joining time. The two pieces form one classical solution beyond T_* , and strong uniqueness identifies it with the history generated by u_0 . This contradicts maximality.

By Theorem C.5.2, $T_* = \infty$ and the normalized pair (u, p) is smooth on every finite interval.

Thus

$$u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)).$$

This proves (4.4.1).

The energy identity proved in Proposition A.2.1 holds on every finite interval of this global solution and gives (4.4.2). In particular,

$$\sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx \leq \int_{\mathbb{R}^3} |u_0(x)|^2 dx < \infty,$$

where the last finiteness follows from the rapid decay in (1.2). Global smoothness of u, p and the uniform kinetic-energy bound are exactly the two conclusions stated in (1.3). $\square$

5 Further Consequences and the Euler Boundary

The finite-time contradiction is complete, but the bounded rectangles carry more information than continuation alone consumes. They show how the same global history distributes vorticity across scales, where the classical criteria sit inside the grid, what becomes of the local singularity picture, and exactly why the argument stops when viscosity is removed.

5.1 A finite sum across infinitely many scales

The finite-time contradiction is already complete at the $(N, M) = (0, 1)$ rectangle, where the L^3 escape is blocked. The next spatial row is not needed for closure; it reveals how much vorticity the same global history can carry across all scales. Fix $T > 0$. On $[0, T]$, the block $\mathcal{R}_{0,2}[u_0; T]$ gives $u \in L^2(0, T; H^3)$, which is strong enough to sum the vorticity carried by every dyadic scale. Let $(\Delta_j)_{j \geq -1}$ be an inhomogeneous Littlewood–Paley decomposition (Bahouri, Chemin, and Danchin 2011, Chapter 2). For $j \geq 0$, the block $\Delta_j \omega$ is concentrated near wavelength 2^{-j} , and $\|\Delta_j \omega\|_\infty$ measures the strongest rotation present in that band. Define

$$\|\omega\|_{B_{\infty,1}^0} := \sum_{j \geq -1} \|\Delta_j \omega\|_\infty.$$

Bernstein’s inequality (Bahouri, Chemin, and Danchin 2011, Chapter 2) gives, for $j \geq 0$,

$$\|\Delta_j \omega\|_\infty \leq C 2^{3j/2} \|\Delta_j \omega\|_2.$$

Insert $2^{3j/2} = 2^{-j/2} 2^{2j}$, sum, and use Cauchy–Schwarz in the shell index:

$$\begin{aligned} \sum_{j \geq 0} \|\Delta_j \omega\|_\infty &\leq C \left(\sum_{j \geq 0} 2^{-j} \right)^{1/2} \left(\sum_{j \geq 0} 2^{4j} \|\Delta_j \omega\|_2^2 \right)^{1/2} \\ &\leq C \|\omega\|_{H^2} \leq C \|u\|_{H^3}. \end{aligned}$$

Bernstein costs $3/2$ derivatives in each high-frequency shell. The H^2 vorticity weight supplies two derivatives, leaving the factor $2^{-j/2}$; its square is summable. That spare half derivative

is what turns one finite Sobolev row into an ℓ^1 sum across all dyadic shells. The low block $\Delta_{-1}\omega$ is controlled by $\|\omega\|_2$, so

$$\|\omega\|_{B_{\infty,1}^0} \leq C \|u\|_{H^3}. \quad (5.1.1)$$

Since $\mathcal{R}_{0,2}[u_0; T]$ gives $u \in L^2(0, T; H^3)$, Cauchy–Schwarz in time yields

$$\sum_{j \geq -1} \int_0^T \|\Delta_j \omega(t)\|_\infty dt = \int_0^T \|\omega(t)\|_{B_{\infty,1}^0} dt < \infty. \quad (5.1.2)$$

What has been made finite is the sum across all shells, not the number of shells that may become active. A recruited core can hand activity from one scale to the next, but the time-integrated shell suprema cannot build a nonsummable cascade before a finite time. On every chosen $[0, T]$, the same low rectangle supplies $L_t^2 H_x^3$ for this shell action and $C_t H_x^2$ for uniform decay of the critical Fourier tail. This is still a finite-order consequence: it does not assert an analytic radius or a weighted sum over every derivative order.

5.2 Three classical criteria inside one low rectangle

The L^3 endpoint first entered as the escape forced by a finite last time. Now fix any finite $T > 0$. The block $\mathcal{R}_{0,1}[u_0; T]$ returns to that same norm from the other direction. Its adjacent pair gives $u \in C([0, T]; H^1)$, and hence

$$\sup_{0 \leq t \leq T} \|u(t)\|_3 < \infty.$$

Thus $\mathcal{R}_{0,1}$ supplies the hypothesis of the Escauriaza–Seregin–Šverák endpoint continuation theorem (Escauriaza, G. A. Seregin, and Šverák 2003). The norm that defined the singular threat is already bounded by the first critical rectangle.

The upper edge of the same rectangle gives a different historical reading:

$$u \in L^2(0, T; H^2) \hookrightarrow L^2(0, T; L^\infty).$$

This is the endpoint pair $(q, p) = (2, \infty)$ on the Ladyzhenskaya–Prodi–Serrin line $2/q + 3/p = 1$ (Prodi 1959; Serrin 1962; Serrin 1963; Ladyzhenskaya 1967). The first meets the exact L^3 norm that defined the terminal threat. The second places the same velocity on the classical scale-balanced integrability line.

There is a third reading closer to the vortex itself. From $u \in L^2(0, T; H^2)$,

$$\nabla u \in L^2(0, T; H^1) \hookrightarrow L^2(0, T; L^3),$$

which supplies the gradient criterion of Beirão da Veiga (Beirão da Veiga 1995). The rectangle that blocks velocity escape also controls the shear responsible for vortex stretching. Together, the three criteria read velocity escape, scale-balanced integrability, and vortex-producing shear from the same low finite block. Repeating the finite-coordinate calculation on the chosen $[0, T]$ gives, for finite $r, k \in \mathbb{N}_0$, $2 \leq p \leq \infty$, and $1 \leq q \leq \infty$,

$$\nabla^k \partial_t^r u \in L^q(0, T; L^p(\mathbb{R}^3)),$$

after choosing a finite Sobolev height $m > k + \frac{3}{2} - \frac{3}{p}$.

5.3 The local singularity picture closes as well

The classical picture in Section A.2 left two descriptions of a possible finite singular time. The critical L^3 norm would have to escape, and every sufficiently small parabolic neighbourhood of a singular point would have to retain a definite scale-invariant velocity–pressure concentration. The finite rectangles exclude both descriptions on the same global history.

Fix $T > 0$. The mixed velocity and pressure estimates give

$$K_T := \sup_{0 \leq t \leq T} (\|u(t)\|_{L^\infty} + \|p(t)\|_{L^\infty}) < \infty. \quad (5.3.1)$$

For a parabolic cylinder

$$Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0) \subset \mathbb{R}^3 \times (0, T],$$

its scale-invariant velocity–pressure content satisfies

$$\rho^{-2} \iint_{Q_\rho(z_0)} (|u|^3 + |p|^{3/2}) \, dx \, dt \leq C \rho^3 (K_T^3 + K_T^{3/2}) \longrightarrow 0 \quad (\rho \downarrow 0). \quad (5.3.2)$$

Thus no spacetime point can retain the positive scale-invariant content required by the Caffarelli–Kohn–Nirenberg singularity picture. At the same time, (4.1.1) rules out the finite-time critical L^3 escape. For the compactly supported data covered by Seregin’s terminal blow-up theorem and Tao’s quantitative refinement (G. Seregin 2012; Tao 2021), the same uniform bound excludes both the divergence and its quantitative lower-rate refinements.

These are not extra hypotheses placed on the solution after global smoothness is known. They are finite projections of the same compatible rectangles that closed the endpoint. On every finite interval the solution has every finite mixed velocity and pressure norm requested above, while Proposition A.2.1 keeps kinetic energy uniform for all time. No uniform-in-time bound for every higher Sobolev norm, analytic radius, or all-orders constant is claimed.

5.4 Why the Euler Tower is different

Now remove viscosity and watch exactly which part of the argument disappears. The mixed derivative jet can still be written for Euler; the spatial ascent that drives this paper cannot. To keep the two histories distinct, write

$$V_n^{\text{NS}} := \partial_t^n u^{\text{NS}}, \quad Q_n^{\text{NS}} := \partial_t^n p^{\text{NS}},$$

and define $V_n^{\text{E}}, Q_n^{\text{E}}$ from a separate Euler solution. Their temporal recurrences are

$$V_{n+1}^{\text{NS}} + \sum_{a=0}^n \binom{n}{a} (V_a^{\text{NS}} \cdot \nabla) V_{n-a}^{\text{NS}} + \nabla Q_n^{\text{NS}} = \nu \Delta V_n^{\text{NS}}, \quad (5.4.1)$$

$$V_{n+1}^{\text{E}} + \sum_{a=0}^n \binom{n}{a} (V_a^{\text{E}} \cdot \nabla) V_{n-a}^{\text{E}} + \nabla Q_n^{\text{E}} = 0. \quad (5.4.2)$$

Both equations retain the differentiated transport and the nonlocal pressure response. Only Navier–Stokes keeps the term $\nu \Delta V_n^{\text{NS}}$, the linear path from temporal row n to two additional spatial derivatives of that same response.

For the two separate histories, set

$$E_s^{\text{NS}} := \frac{1}{2} \left\| \Lambda^s u^{\text{NS}} \right\|_2^2, \quad E_s^{\text{E}} := \frac{1}{2} \left\| \Lambda^s u^{\text{E}} \right\|_2^2,$$

with corresponding transfer terms $\mathcal{P}_s^{\text{NS}}$ and $\mathcal{P}_s^{\text{E}}$. Then

$$(E_s^{\text{NS}})' = \mathcal{P}_s^{\text{NS}} - 2\nu E_{s+1}^{\text{NS}}, \quad (E_s^{\text{E}})' = \mathcal{P}_s^{\text{E}}. \quad (5.4.3)$$

For Navier–Stokes, repeated differentiation and division by $(-2\nu)^n$ isolate $E_{n+1/2}^{\text{NS}}$ in (2.3.6). Euler retains the nonlinear mixing of time and space, but it has no signed coercive rung that isolates the next Sobolev height in this way.

An Euler history lying in an all-orders compatible intersection would satisfy every fixed Sobolev bound and, in particular, the Beale–Kato–Majda condition $\int_0^T \|\omega^{\text{E}}(t)\|_\infty dt < \infty$ (Beale, Kato, and Majda 1984). The fixed-viscosity theorem used here does not supply that Euler intersection, and its constants are not claimed to remain bounded as $\nu \downarrow 0$. This is the exact boundary of the method. Euler keeps the mixed jet and all of its nonlinear coupling. What it loses is the signed viscous rung that lets the completed row isolate the next Sobolev height. For fixed positive viscosity, the rectangle theorem bounds the finite coordinates exposed by that ascent on any putative terminal interval. This rules out a finite last time. On the resulting global history, every finite mixed coordinate is bounded on each finite interval, and Proposition A.2.1 keeps kinetic energy uniformly bounded for all time.

A Local Construction and Classical Estimates

The main argument uses local theory twice: first to construct the maximal history, and then to continue it from a smooth terminal value. We prove the required local theorem here and then set out the classical estimates that locate the finite-endpoint obstruction.

A.1 The local construction behind the maximal history

Local existence and restart use one construction. Its lifespan depends only on the fixed viscosity and the H^3 size of the datum; once that interval is fixed, every higher Sobolev order already present in the datum persists on it.

For real s , write

$$H_\sigma^s(\mathbb{R}^3) := \{v \in H^s(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

Proposition A.1.1 (Local construction and continuation alternative). *Let $t_0 \geq 0$, $\nu > 0$, and $a \in H_\sigma^3(\mathbb{R}^3)$. There is a time $\delta = \delta(\nu, \|a\|_{H^3}) > 0$ and a unique normalized-pressure solution (w, q) with*

$$w \in C([t_0, t_0 + \delta]; H_\sigma^3) \cap L^2(t_0, t_0 + \delta; H^4), \quad w_t \in L^2(t_0, t_0 + \delta; H^2).$$

If $a \in H_\sigma^m$ for an integer $m > 3$, the same interval also carries

$$w \in C([t_0, t_0 + \delta]; H_\sigma^m) \cap L^2(t_0, t_0 + \delta; H^{m+1}), \quad w_t \in L^2(t_0, t_0 + \delta; H^{m-1}).$$

The local solutions join uniquely into one maximal history, and a finite maximal time forces its H^3 norm to become unbounded along the terminal approach.

Proof of Proposition A.1.1. Fix $t_0 \geq 0$, $\nu > 0$, and $a \in H_\sigma^3(\mathbb{R}^3)$. For real s , we use the inhomogeneous norm

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} |\hat{f}(\xi)|^2 d\xi, \quad \langle \xi \rangle = (1 + |\xi|^2)^{1/2}.$$

The Leray projection is denoted by $\mathbb{P}$, and R_j denotes the Riesz transform with symbol $i\xi_j/|\xi|$. Both are bounded on every H^s .

Why the base class has exactly these three rows

The nonlinear term fixes the base spatial index. In three dimensions,

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3), \quad \|fg\|_{H^3} \leq C \|f\|_{H^3} \|g\|_{H^3}.$$

Thus an H^3 velocity is Lipschitz and H^3 is an algebra. In particular,

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^2} \leq C \|f\|_{H^3} \|g\|_{H^3}. \quad (\text{A.1.1})$$

For integer derivatives this follows directly from Leibniz' rule and the Sobolev embeddings $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$; density gives the stated estimate on H^3 .

The heat operator determines the two neighbouring rows. A force in H^2 produces one square-integrable parabolic derivative above the base state, $z \in L_t^2 H_x^4$, and the equation places z_t back in $L_t^2 H_x^2$. These rows meet exactly at H^3 . Indeed, whenever

$$v \in L^2(J; H^4), \quad v_t \in L^2(J; H^2),$$

frequency truncation, the fundamental theorem of calculus, and Cauchy–Schwarz give

$$\left| \|v(t)\|_{H^3}^2 - \|v(r)\|_{H^3}^2 \right| \leq 2 \|v_t\|_{L^2(r,t;H^2)} \|v\|_{L^2(r,t;H^4)}. \quad (\text{A.1.2})$$

The right side tends to zero on shrinking time intervals. Passing from the frequency truncations shows that v has a unique representative in $C(J; H^3)$. This is why the local strong class is

$$C_t H_x^3 \cap L_t^2 H_x^4, \quad \partial_t v \in L_t^2 H_x^2.$$

The parabolic gain $v \in L_t^2 H_x^4$ and the equation row $v_t \in L_t^2 H_x^2$ meet at the continuous H^3 trace needed to carry the initial value.

The shifted linear estimate

Let $J_\delta = [t_0, t_0 + \delta]$, where $0 < \delta \leq 1$, and consider

$$z_t - \nu \Delta z = F, \quad z(t_0) = a, \quad F \in L^2(J_\delta; H^2).$$

The H^3 energy pairs the force with z through the adjacent spaces H^2 and H^4 . In Fourier variables define

$$\langle F, z \rangle_{H^2, H^4; H^3} := \int_{\mathbb{R}^3} \langle \xi \rangle^6 \widehat{F}(\xi) \cdot \overline{\widehat{z}(\xi)} d\xi.$$

The factorization $\langle \xi \rangle^6 = \langle \xi \rangle^2 \langle \xi \rangle^4$ and Cauchy–Schwarz give

$$|\langle F, z \rangle_{H^2, H^4; H^3}| \leq \|F\|_{H^2} \|z\|_{H^4}. \quad (\text{A.1.3})$$

For a smooth solution, the H^3 energy identity is

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \text{Re} \langle F, z \rangle_{H^2, H^4; H^3}.$$

The inhomogeneous weights also give the exact splitting

$$\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2.$$

Using (A.1.3), Young’s inequality, and then Gronwall’s inequality yields

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right).$$

Since $z_t = \nu \Delta z + F$, the same estimate also controls the time derivative:

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} + \|z_t\|_{L^2(J_\delta; H^2)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right). \quad (\text{A.1.4})$$

The constant is independent of $\delta \in (0, 1]$. Apply a bounded frequency cutoff to a and F , solve the truncated equations, and use (A.1.4) on differences. The limits solve the general linear problem, and (A.1.2) supplies the prescribed strong H^3 trace at t_0 . The same estimate applied to a difference gives linear uniqueness.

The projected fixed point

Set

$$X_\delta := C(J_\delta; H_\sigma^3) \cap L^2(J_\delta; H^4), \quad \|w\|_{X_\delta} := \|w\|_{CH^3} + \|w\|_{L^2H^4}.$$

The projected mild equation is

$$\Phi(w)(t) := e^{\nu(t-t_0)\Delta}a - \int_{t_0}^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (w \otimes w)(s) ds. \quad (\text{A.1.5})$$

All abbreviated time norms below are taken over J_δ . Estimate (A.1.1) and the length of the interval give

$$\|\mathbb{P} \nabla \cdot (w \otimes w)\|_{L^2H^2} \leq C\delta^{1/2} \|w\|_{CH^3}^2, \quad (\text{A.1.6})$$

$$\|\mathbb{P} \nabla \cdot (w \otimes w - v \otimes v)\|_{L^2H^2} \leq C\delta^{1/2} (\|w\|_{CH^3} + \|v\|_{CH^3}) \|w - v\|_{CH^3}. \quad (\text{A.1.7})$$

Inserting these bounds into (A.1.4) gives

$$\|\Phi(w)\|_{X_\delta} \leq C_\nu \|a\|_{H^3} + C_\nu C\delta^{1/2} \|w\|_{X_\delta}^2$$

and

$$\|\Phi(w) - \Phi(v)\|_{X_\delta} \leq C_\nu C\delta^{1/2} (\|w\|_{X_\delta} + \|v\|_{X_\delta}) \|w - v\|_{X_\delta}.$$

For $a \neq 0$, take $R = 2C_\nu \|a\|_{H^3}$ and choose $\delta \leq 1$ so that $4C_\nu C\delta^{1/2} R \leq 1$. The map Φ then preserves the closed X_δ -ball of radius R and has Lipschitz constant at most $1/2$ there. If $a = 0$, the fixed point is the zero solution. Banach's fixed-point theorem produces $w \in X_\delta$, and the choice may be made so that

$$\delta \geq \min \left\{ 1, c_\nu (1 + \|a\|_{H^3})^{-2} \right\}. \quad (\text{A.1.8})$$

Here $c_\nu > 0$ depends only on the fixed viscosity and the normalization of the Sobolev estimates.

The contraction first proves uniqueness inside the chosen ball. It also gives uniqueness in the full stated strong class. If $w, v \in X_\delta$ have the same datum, apply (A.1.4) and (A.1.7) to $w - v$ on a sufficiently short initial subinterval. The coefficient is then smaller than one, so $w = v$ there. Their common endpoint is the initial value for the next subinterval. A finite partition of J_δ repeats the argument across the entire interval. The heat semigroup and $\mathbb{P}$ preserve divergence-free fields, and the projected equation together with (A.1.6) gives

$$w_t \in L^2(J_\delta; H^2). \quad (\text{A.1.9})$$

Recovering the normalized pressure

Define

$$q := R_i R_j (w_i w_j). \quad (\text{A.1.10})$$

Because H^3 is an algebra and the Riesz transforms are bounded on H^3 ,

$$q \in C(J_\delta; H^3).$$

For $F = \nabla \cdot (w \otimes w)$, the Fourier symbols give $\mathbb{P}F = F + \nabla q$. The projected fixed point consequently solves

$$w_t + (w \cdot \nabla)w + \nabla q = \nu \Delta w, \quad \nabla \cdot w = 0, \quad w(t_0) = a.$$

The condition $q(t) \in H^3(\mathbb{R}^3)$ fixes the whole-space pressure constant. If two such pressures accompany the same velocity, their difference has zero gradient and belongs to L^2 , hence vanishes. The projected uniqueness just proved is thus uniqueness of the pressure-complete pair (w, q) .

Higher spatial orders on the base interval

Fix an integer $m \geq 3$ and first suppose a is smooth. Write the H^m energy as the sum of the L^2 energies of $\partial^\alpha w$, $|\alpha| \leq m$. Apply ∂^α to the projected equation and pair with $\partial^\alpha w$. The divergence-free transport of the top derivative cancels:

$$\left((w \cdot \nabla) \partial^\alpha w, \partial^\alpha w \right)_2 = 0.$$

The remaining terms in the Leibniz expansion contain at least one derivative on the transporting factor. Summing over $|\alpha| \leq m$, the Sobolev embeddings $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$ give the integer commutator bound

$$\left(\sum_{|\alpha| \leq m} \|[\partial^\alpha, w \cdot \nabla]w\|_2^2 \right)^{1/2} \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}.$$

The differentiated energy identity gives

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}^2. \quad (\text{A.1.11})$$

The base solution lies in $C(J_\delta; H^3)$, so $\|\nabla w\|_\infty$ is bounded, and in particular integrable, on this same interval. Gronwall gives

$$\|w(t)\|_{H^m}^2 \leq \|a\|_{H^m}^2 \exp \left(2C_m \int_{t_0}^t \|\nabla w(s)\|_\infty ds \right). \quad (\text{A.1.12})$$

Keeping the dissipative term in (A.1.11) gives $w \in L^2(J_\delta; H^{m+1})$. The product estimate at order $m-1$ puts the projected transport term in $L^2(J_\delta; H^{m-1})$, and the equation gives

$$w_t \in L^2(J_\delta; H^{m-1}).$$

The order- m version of (A.1.2) now yields

$$w \in C(J_\delta; H^m) \cap L^2(J_\delta; H^{m+1}), \quad w_t \in L^2(J_\delta; H^{m-1}). \quad (\text{A.1.13})$$

Formula (A.1.10) and the H^m algebra estimate give $q \in C(J_\delta; H^m)$.

For a general $a \in H_\sigma^m$, approximate a by divergence-free frequency cutoffs a_k . Their H^3 norms are bounded by a common multiple of $\|a\|_{H^3}$, so (A.1.8) places every smooth solution on the same base interval. The fixed-point estimate identifies their limit with the H^3 solution already constructed, while (A.1.11)–(A.1.12) give uniform bounds in the three spaces of (A.1.13). Weak compactness identifies the higher-order limits with w . The adjacent trace gives a continuous H^m representative, and the energy inequality at t_0 , combined with weak convergence to a , gives strong convergence $w(t) \rightarrow a$ in H^m as $t \downarrow t_0$.

No higher order changes the interval: its length was chosen once from ν and $\|a\|_{H^3}$. If $a \in H_\sigma^\infty$, applying (A.1.13) at each fixed m and using H^3 uniqueness shows that all orders belong to the same pair. Repeatedly differentiating the pressure formula and the momentum equation then gives smoothness in time with values in every finite Sobolev space. Spatial Sobolev embedding makes (w, q) smooth in (x, t) on the construction interval.

The maximal history and its continuation alternative

Starting at t_0 , join all local intervals furnished by the construction. Strong uniqueness identifies the solutions and normalized pressures on every overlap, so their union is one pressure-complete solution on a maximal interval $[t_0, T_{\max})$.

Suppose $T_{\max} < \infty$ and

$$\sup_{t_0 \leq t < T_{\max}} \|w(t)\|_{H^3} \leq M.$$

The lifespan calculation supplies the same positive lower bound

$$\delta_M := \min \{1, c_\nu(1 + M)^{-2}\}$$

whenever the construction is started from the actual datum $w(t)$ at a later time. Choose $t > T_{\max} - \delta_M/2$. The solution starting from $w(t)$ exists beyond $T_{\max}$, and uniqueness identifies it with the maximal history on their common interval. This extends that history past its maximal time, which is impossible. We have proved the continuation alternative

$$T_{\max} < \infty \implies \limsup_{t \uparrow T_{\max}} \|w(t)\|_{H^3} = \infty. \quad (\text{A.1.14})$$

Every higher order present in the initial datum persists on each local piece, because each piece uses the same H^3 -controlled lifespan. For $a \in \mathcal{S}_\sigma \subset H_\sigma^\infty$, this gives the smooth pressure-normalized maximal history asserted in the proposition. $\square$

A.2 The Classical Estimates Before Closure

Local theory gives the smooth fluid a beginning, so the first serious question is what the equations preserve after that beginning. The first answer is kinetic energy, and almost every later regularity result can be read as an attempt to recover the derivative control that energy leaves behind.

Energy and the First Uncontrolled Derivative

Proposition A.2.1 (Classical kinetic-energy identity). *Let (u, p) be a smooth solution of (1.1) with sufficient spatial decay. Then, for every time t in its interval of existence,*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds = \|u_0\|_{L^2}^2. \quad (\text{A.2.1})$$

Proof. Multiply the momentum equation by u and integrate over space. Incompressibility removes nonlinear transport from the total energy because

$$\int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u \, dx = \frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 \, dx = 0.$$

It removes the pressure contribution for the same divergence-free reason:

$$\int_{\mathbb{R}^3} \nabla p \cdot u \, dx = - \int_{\mathbb{R}^3} p \nabla \cdot u \, dx = 0.$$

Viscosity is the surviving term, and integration by parts gives

$$\nu \int_{\mathbb{R}^3} \Delta u \cdot u \, dx = -\nu \|\nabla u\|_{L^2}^2.$$

Integrating the resulting equality in time proves (A.2.1). □

The velocity has an L^2 ceiling at every time, and viscosity pays for the squared L^2 gradient after it is accumulated in time.

Leray found a way to keep this estimate when smoothness is no longer assumed. Regularization, compactness, and the energy inequality produce a global weak solution satisfying (Leray 1934)

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; H^1(\mathbb{R}^3))$$

on every finite interval. This is genuine global existence at the energy level; the energy inequality alone does not settle uniqueness or spontaneous regularity for arbitrary three-dimensional Leray solutions.

The Sobolev embedding

$$H^1(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3)$$

turns Leray's spatial derivative into

$$u \in L^2(0, T; L^6(\mathbb{R}^3)).$$

That extra integrability is immediately spent on the nonlinear transport. For example,

$$\|(u \cdot \nabla) u\|_{L^{3/2}} \leq \|u\|_{L^6} \|\nabla u\|_{L^2},$$

so the transport term is integrable in time with values in $L^{3/2}$. This is strong enough to keep the equation meaningful in the weak construction. It still supplies no time-uniform ceiling on $\|\nabla u(t)\|_{L^2}$, because a sequence of higher and shorter peaks can have a finite squared time integral.

Vorticity shows why that opening is specifically dangerous in three dimensions. Set

$$\omega := \nabla \times u.$$

For the divergence-free decaying fields considered here,

$$\|\nabla u\|_{L^2}^2 = \|\omega\|_{L^2}^2,$$

so the dissipated gradient is also the enstrophy. Taking the curl of the momentum equation gives

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The new term $(\omega \cdot \nabla) u$ stretches and tilts vorticity with the velocity gradient generated by the same flow. Its energy balance is

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 = \int_{\mathbb{R}^3} (\omega \cdot \nabla) u \cdot \omega \, dx.$$

The stretching integral has no fixed sign, and the energy identity does not force it to remain below viscous dissipation at each time. The first estimate reaches the whole time interval at a level below the derivative control needed to restart the smooth solution.

What Is Strong Enough to Continue the Solution

The natural way to ask for several derivative levels at once is a Sobolev norm. For an integer $m \geq 0$, write

$$\|u\|_{H^m}^2 := \sum_{|\alpha| \leq m} \|\partial^\alpha u\|_{L^2}^2,$$

with the usual Fourier definition when the order is fractional. In three dimensions, Sobolev embedding gives (Cerdà 2010, Theorem 7.28)

$$H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3) \quad \text{when} \quad s > k + \frac{3}{2}.$$

In particular,

$$s > \frac{5}{2} \quad \implies \quad H^s(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3).$$

This is the first level at which one Sobolev norm directly controls the pointwise velocity gradient that multiplies every higher differentiated energy estimate.

Suppose a smooth solution on $[0, T)$ satisfies

$$\sup_{0 \leq t < T} \|u(t)\|_{H^s} \leq M \quad \text{for some } s > \frac{5}{2}.$$

For each fixed finite $m \geq s$, differentiate the equation spatially through order m , pair with the corresponding derivative of u , and sum the resulting identities. The product rule and incompressibility give the standard estimate

$$\frac{1}{2} \frac{d}{dt} \|u\|_{H^m}^2 + \nu \|\nabla u\|_{H^m}^2 \leq C_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2.$$

The assumed H^s ceiling controls the coefficient on the right, and Gronwall's inequality propagates the chosen H^m norm from the smooth datum. The constant may depend on m , because smoothness asks for every finite order in turn rather than one constant for the entire infinite collection.

Pressure does not add an independent unknown at the next stage. Taking the divergence of the momentum equation gives

$$-\Delta p = \sum_{i,j=1}^3 \partial_i \partial_j (u_i u_j),$$

with decay fixing the pressure normalization. Once the spatial derivatives of u are controlled, this elliptic relation recovers the corresponding spatial derivatives of p , and

$$\partial_t u = \nu \Delta u - (u \cdot \nabla) u - \nabla p$$

then recovers the first time derivative. Repeating the same two operations recovers each higher mixed spatial and temporal derivative. One uniform Sobolev bound above $5/2$ has enough leverage to return the complete smooth velocity–pressure pair.

It also gives the missing time needed to cross an endpoint. Local well-posedness in H^s supplies a positive lifespan determined by s , ν , and the size of the initial H^s norm (Fujita and Kato 1964; Kato 1984). Choose $t_0 < T$, regard $u(\cdot, t_0)$ as a fresh initial slice, and restart the local theory from that slice. If $\|u(\cdot, t_0)\|_{H^s} \leq M$ uniformly, the restarted lifespans have a common positive lower bound, so a sufficiently late t_0 carries the same solution past T . Uniqueness on the overlap joins the restarted piece to the original motion. Hence a finite maximal time forces

$$\limsup_{t \uparrow T_*} \|u(t)\|_{H^s} = \infty \quad \text{for every } s > \frac{5}{2}.$$

The distance from Leray's estimate to continuation is now exact: time-integrated H^1 control must somehow become time-uniform control at a Sobolev level above $5/2$.

Critical Scale and Singular Concentration

The equations themselves show where such a conversion can lose information. If (u, p) is a solution, then

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

has the same equation at a different spatial and temporal scale. A quantity left unchanged by this transformation sees an intensifying structure with the same strength no matter how small that structure becomes.

The Ladyzhenskaya–Prodi–Serrin criteria locate exactly this balance between spatial intensity and temporal persistence (Prodi 1959; Serrin 1962; Serrin 1963; Ladyzhenskaya 1967). They give regularity through T when

$$u \in L^q(0, T; L^p(\mathbb{R}^3)), \quad \frac{2}{q} + \frac{3}{p} \leq 1, \quad p > 3.$$

The spatial exponent measures how tightly velocity may concentrate, and the time exponent measures how long that concentration may persist. Equality in the exponent relation is unchanged by Navier–Stokes scaling.

The spatial endpoint of this family is $L_t^\infty L_x^3$, because

$$\|u_\lambda(t)\|_{L^3} = \|u(\lambda^2 t)\|_{L^3}.$$

Escauriaza, Seregin, and Šverák proved that

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3))$$

still carries a solution regularly through T (Escauriaza, G. A. Seregin, and Šverák 2003). Their rescaling argument turns a first singularity into an ancient limiting solution and uses backward uniqueness to rule that limit out. Thus a finite endpoint cannot hide behind a bounded critical L^3 norm.

The local theory of singular concentration reaches a complementary conclusion. For $z_0 = (x_0, t_0)$, write the parabolic cylinder

$$Q_r(z_0) := B_r(x_0) \times (t_0 - r^2, t_0).$$

Caffarelli, Kohn, and Nirenberg proved that sufficiently small scale-invariant velocity and pressure quantities on $Q_r(z_0)$ force regularity on a smaller cylinder (Caffarelli, Kohn, and Nirenberg 1982). A singular point must retain a definite scale-invariant concentration in every sufficiently small parabolic neighbourhood. Their covering argument confines the possible singular set to zero one-dimensional parabolic Hausdorff measure, but a single point would still end global smoothness.

The endpoint result also has a direct blow-up consequence. For the smooth compactly supported whole-space data in Seregin’s theorem, a finite maximal time satisfies (G. Seregin 2012)

$$\lim_{t \uparrow T_*} \|u(t)\|_{L^3(\mathbb{R}^3)} = \infty.$$

Tao’s quantitative refinement forces at least a positive power of a triple logarithm of the inverse time remaining along infinitely many times approaching the endpoint (Tao 2021). That forced rate is extremely slow compared with any power law, so much faster concentration remains compatible with all known critical lower bounds.

What the Classical Estimates Leave Open

At a finite maximal time, kinetic energy stays bounded while every continuation-grade Sobolev norm and the critical L^3 norm lose finite control. A singular point must also retain scale-invariant concentration at arbitrarily small parabolic scales. These are necessary descriptions of the concentration; none supplies the terminal-uniform estimate that would stop it.

The reason can be seen directly from scaling:

$$\|u_\lambda(t)\|_{L^3} = \|u(\lambda^2 t)\|_{L^3}, \quad \|u_\lambda(t)\|_{L^2}^2 = \lambda^{-1} \|u(\lambda^2 t)\|_{L^2}^2.$$

Moving a structure to a shorter length scale preserves its critical L^3 reading while reducing the L^2 energy carried by the rescaled piece. The global energy ceiling cannot become a scale-independent ceiling on that critical concentration.

Scaling also fixes the time available at each stage. If the active length is halved repeatedly, the corresponding parabolic durations are proportional to

$$1, \frac{1}{4}, \frac{1}{16}, \dots,$$

and the entire schedule fits inside a finite total time. The classical estimates above do not exclude this schedule: each finer stage can occupy less time while retaining critical size. This is the precise role of the Zeno picture here. It records the timing left open by the estimates; it does not construct a singular Navier–Stokes fluid.

The estimate still missing from the classical picture is a terminal-uniform bound in one continuation-grade norm.

B Derivative and Finite-Record Calculus

Time differentiation opens the temporal rows, and Navier–Stokes identifies their transport, pressure, and viscosity content. The finite-time lemma then propagates escape through those existing rows under its stated hypotheses. Spatial differentiation opens the corresponding columns, and adjacent Sobolev pairs give those rows a common time trace.

B.1 Critical Height at a Finite Endpoint

Fix a rapidly decaying datum u_0 on $\mathbb{R}^3$, a viscosity $\nu > 0$, and the maximal classical solution (u, p) they generate on $[0, T_*)$. Suppose $T_* < \infty$. The field is smooth at every earlier time, so any concentration has to be followed through smaller scales as t approaches that endpoint.

Kinetic energy stays bounded along the classical interval, but a fixed amount of energy is not equally visible at every scale. The Navier–Stokes scaling shows the difference. For every $\lambda > 0$, the rescaled fields

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

solve the same equation with the same viscosity on the rescaled time interval. Length has been divided by λ , time by λ^2 , and velocity has been multiplied by λ . This is the parabolic relation already present in the equation: one time derivative and the two spatial derivatives in viscosity acquire the same factor under rescaling.

Let $\Lambda = (-\Delta)^{1/2}$, and measure a spatial Sobolev height by

$$\|v\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{v}(\xi)|^2 d\xi.$$

Proposition B.1.1 (Critical Sobolev scaling). *For every $s \in \mathbb{R}$ for which the homogeneous norm is finite and every $\lambda > 0$,*

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2.$$

In particular, $s = \frac{1}{2}$ is the unique scale-invariant Sobolev height.

Proof. The Fourier transform of the rescaled velocity is

$$\widehat{u}_\lambda(\xi, t) = \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t).$$

Putting this into the Sobolev norm and changing variables from ξ to $\eta = \xi/\lambda$ gives

$$\begin{aligned} \|u_\lambda(t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} \left| \widehat{u}(\xi/\lambda, \lambda^2 t) \right|^2 d\xi \\ &= \lambda^{2s-1} \int_{\mathbb{R}^3} |\eta|^{2s} \left| \widehat{u}(\eta, \lambda^2 t) \right|^2 d\eta. \end{aligned}$$

The exponent $2s - 1$ vanishes exactly at $s = \frac{1}{2}$, which proves the proposition. $\square$

Every Sobolev height tilts under the compression except $s = 1/2$. At that height the power of λ vanishes, so define

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2.$$

This is the critical height: exact rescaling can move a smooth configuration to a finer length and a shorter time without changing the value of H .

The kinetic-energy scale moves in the other direction. Putting $s = 0$ in the same calculation gives

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \left\| u(\lambda^2 t) \right\|_2^2.$$

A finer rescaled structure can carry less L^2 energy while standing at the same critical height. That is the opening left by the ordinary energy estimate: it can keep the total energy finite while the scale at which the fluid is changing continues to fall.

The finite parabolic schedule in Section A.2 shows that progressively smaller scale durations can fit before one finite time. At the height just derived, each finer copy can keep the same value of H while its L^2 contribution shrinks. Scaling alone does not make the actual fluid pass through those copies. The equation would have to generate each finer state from the state before it while transport, pressure, incompressibility, and viscosity all continue to act.

Critical height fixes the scale. The finite endpoint asks how fast the field can move through an unbounded sequence of regular states, which is a question about one time response and the response immediately above it.

B.2 What finite time implies after one response escapes

The temporal Tower is already present once the same velocity field is differentiated in time. Finite time adds a separate implication after one of those responses is known to escape: an integrable next response would give the escaping row a finite terminal value. The argument below propagates an escape through existing rows under its stated hypotheses; it does not produce those rows.

Fix one Banach space B , retain the Eulerian rows

$$V_n(t) := \partial_t^n u(\cdot, t),$$

and suppose that $V_n : [0, T_*) \rightarrow B$ is locally absolutely continuous, with Bochner derivative V_{n+1} . For $0 \leq s < t < T < T_*$, the Banach-valued fundamental theorem of calculus gives

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau) d\tau$$

and

$$\|V_n(t) - V_n(s)\|_B \leq \int_s^t \|V_{n+1}(\tau)\|_B d\tau.$$

If

$$\sup_{0 < t < T_*} \|V_n(t)\|_B = \infty,$$

then the finiteness of $\|V_n(0)\|_B$ forces

$$\int_0^{T_*} \|V_{n+1}(\tau)\|_B d\tau = \infty.$$

Because $T_* < \infty$, this in turn forces

$$\sup_{0 < t < T_*} \|V_{n+1}(t)\|_B = \infty.$$

This iteration is valid only while the same fixed space B , absolute continuity, and the Bochner-derivative identity hold at every successive row. That is the precise statement of finite-time response escalation for the temporal rows used in the paper.

B.3 The Pressure-Complete Mixed Jet

At temporal order n , the recurrences (2.2.2) and (2.2.4) contain V_{n+1} , $\nu \Delta V_n$, every binomial split of the transport term, and the normalized pressure row Q_n . To see the spatial derivatives already present in that equation, begin with one derivative of a transport factor. Differentiate one transport factor in the coordinate direction x_ℓ :

$$\partial_\ell ((V_a \cdot \nabla) V_{n-a}) = ((\partial_\ell V_a) \cdot \nabla) V_{n-a} + (V_a \cdot \nabla) \partial_\ell V_{n-a}.$$

At higher spatial order, every derivative divides between the two factors. For $\alpha \in \mathbb{N}_0^3$, write the corresponding velocity and pressure derivatives as

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p,$$

and for $\beta \leq \alpha$, set

$$\binom{\alpha}{\beta} := \prod_{k=1}^3 \binom{\alpha_k}{\beta_k}.$$

The simultaneous temporal and spatial product rules give

$$\partial_t^n \partial_x^\alpha ((u \cdot \nabla)u) = \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta}.$$

Substitution into the equation exposes the full mixed recurrence:

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} \\ + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned}$$

Its pressure coordinates satisfy

$$-\Delta \Pi_{n,\alpha} = \partial_i \partial_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta})_i (J_{n-a,\alpha-\beta})_j.$$

Every $J_{n,\alpha}$ and $\Pi_{n,\alpha}$ above is a derivative of the original pressure-complete solution. A finite rectangle selects a finite range of n and α from this one derivative family; it does not introduce a second solution.

The viscous term explains why moving upward in n also reaches farther into the spatial derivatives: $J_{n+1,\alpha}$ is coupled to $\Delta J_{n,\alpha}$. The critical-height calculation now isolates that spatial gain without discarding the nonlinear and pressure terms beside it.

B.4 The Spatial Column Inside Critical Height

To see exactly how one time derivative reaches into space, take a scalar energy from one spatial level of the mixed jet. For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2,$$

so that

$$H(t) = E_{1/2}(t).$$

Keeping s free before returning to $1/2$ exposes the mechanism that repeats at every height. Apply Λ^s to the pressure-complete momentum equation and pair it in L^2 with $\Lambda^s u$. The temporal term gives

$$\langle \Lambda^s u, \Lambda^s \partial_t u \rangle = \frac{d}{dt} \frac{1}{2} \|\Lambda^s u\|_2^2 = E'_s.$$

Rapid decay and commutation with spatial derivatives give

$$\nu \langle \Lambda^s u, \Lambda^s \Delta u \rangle = -\nu \|\Lambda^{s+1} u\|_2^2 = -2\nu E_{s+1}.$$

The pressure pairing vanishes only after incompressibility and integration by parts have done their work:

$$\langle \Lambda^s u, \Lambda^s \nabla p \rangle = -\langle \nabla \cdot \Lambda^s u, \Lambda^s p \rangle = 0.$$

This global cancellation does not remove pressure from the local equation or from the mixed jet that generated the balance. Keep the signed nonlinear transfer in its pressure-complete form,

$$\mathcal{P}_s(t) := - \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle.$$

The exact balance is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

Thus one time derivative of E_s introduces the next spatial energy E_{s+1} together with the complete signed transfer at height s .

At critical height, the first row reads

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

This is already the first spatial ascent, from the scale-invariant $E_{1/2}$ to $E_{3/2}$. It is not a bound for $E_{3/2}$, because the transfer and dissipative terms can cancel inside the complete value of H' . The whole row has to be differentiated before its next spatial consequence can be seen.

The balance one step higher is

$$E'_{3/2} = \mathcal{P}_{3/2} - 2\nu E_{5/2}.$$

Substituting it into the derivative of the first critical row gives

$$\begin{aligned} H'' &= \mathcal{P}'_{1/2} - 2\nu E'_{3/2} \\ &= \mathcal{P}'_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

The second derivative now contains $E_{5/2}$ and the transfers from both heights $1/2$ and $3/2$. Differentiating once more uses

$$E'_{5/2} = \mathcal{P}_{5/2} - 2\nu E_{7/2}$$

and yields

$$H''' = \mathcal{P}''_{1/2} - 2\nu \mathcal{P}'_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

One more response gives

$$H^{(4)} = \mathcal{P}'''_{1/2} - 2\nu \mathcal{P}''_{3/2} + (2\nu)^2 \mathcal{P}'_{5/2} - (2\nu)^3 \mathcal{P}_{7/2} + (2\nu)^4 E_{9/2}.$$

After each time differentiation, the highest spatial energy is one step higher and every earlier transfer is still present.

The pattern is triangular rather than diagonal.

Proposition B.4.1 (Critical-height spatial ascent). *For every integer $n \geq 1$,*

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

Proof. The critical balance gives the case $n = 1$. Assume the identity at order n , differentiate it, and substitute

$$E'_{n+1/2} = \mathcal{P}_{n+1/2} - 2\nu E_{n+3/2}.$$

The substitution produces the next transfer and $E_{n+3/2}$ with the required coefficient, which proves the identity by induction. $\square$

The identity still supplies no estimate by itself. Even if $H^{(n)}$ is known, $E_{n+1/2}$ cannot be isolated without controlling the entire signed transfer sum. Its appearance in the formula does not bound it.

The homogeneous scale made the critical dilation exact, while continuation is usually read in the inhomogeneous scale. For $s > 0$,

$$\|u(t)\|_{H^s}^2 \simeq \|u(t)\|_2^2 + \|\Lambda^s u(t)\|_2^2 = \|u(t)\|_2^2 + 2E_s(t).$$

The unforced energy identity supplies the missing low-frequency part:

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2.$$

Thus control of $E_s(t)$ supplies the high-frequency part of an H^s bound, and an L^1 -in-time bound for E_s supplies the high-frequency part of an $L_t^2 H_x^s$ bound on each finite interval.

The calculation begins at $1/2$ because that height is fixed by scaling, and its first derivative reaches $3/2$. At the endpoint, the relation runs in the other direction: the whole-terminal Sobolev rows first control the $1/2$ and $3/2$ levels, and those levels then control H and its first dissipative height.

Any finite number of time derivatives reaches only a finite spatial height. To retain every finite height required by the full derivative family, the same velocity must belong to every finite temporal and spatial Sobolev row.

B.5 Simultaneous Sobolev Intersections

The critical-height calculation reaches spatial energies of arbitrarily high order, but any one derivative request stops at a finite order. Suppose, for example, that one wants $\partial_x^\alpha u$ to be continuous for every $|\alpha| \leq k$. In three dimensions, the Sobolev embedding

$$H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3)$$

holds whenever

$$m > k + \frac{3}{2}.$$

One requested spatial order k comes from one finite height m above its embedding threshold. As k ranges through all finite orders, no last m can be chosen, but no infinite-order norm is required either. The exact spatial object is the coordinatewise intersection

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3),$$

whose membership means that every finite Sobolev norm is finite on the same field. For each fixed k , one of those finite memberships yields the desired C^k control, so

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Finite-time response asks for the same construction in the temporal direction. Let $I = (0, T)$ with $T < T_*$, and let X be a Hilbert space. The vector-valued embedding

$$H^r(I; X) \hookrightarrow C^q(\bar{I}; X)$$

holds whenever

$$r > q + \frac{1}{2}.$$

One requested time derivative of order q comes from one finite temporal height r . Allowing every finite request gives

$$\bigcap_{r=0}^{\infty} H^r(I; X) \hookrightarrow C^\infty(\bar{I}; X).$$

The time intersection is again coordinatewise: each q chooses its own finite r .

The mixed jet requires these choices at the same time. Ask for one concrete coordinate,

$$\partial_t^2 \partial_x^\alpha u, \quad |\alpha| \leq 4.$$

The single finite membership

$$u \in H^3(I; H^6(\mathbb{R}^3))$$

places that coordinate in

$$C^2(\bar{I}; C^4(\mathbb{R}^3)),$$

because

$$3 > 2 + \frac{1}{2}, \quad 6 > 4 + \frac{3}{2}.$$

Nothing special happened at the pair $(2, 4)$. For arbitrary finite temporal and spatial orders q and k , choose finite indices satisfying

$$r > q + \frac{1}{2}, \quad m > k + \frac{3}{2}.$$

Then

$$H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

To satisfy every finite time-space request on the same velocity, take the intersection of these finite Sobolev memberships:

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)).$$

Membership means

$$u \in \mathcal{V}_\infty(I) \iff \|u\|_{H^r(I; H^m)} < \infty \text{ for every finite pair } (r, m).$$

Each membership is a coordinate of the same u , and the intersection keeps their norms separate. The embeddings above then give

$$\mathcal{V}_\infty(I) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3).$$

Choose temporal and spatial indices large enough to give a continuous trace above the critical levels. That trace contains both

$$u(t) \in H^{1/2}(\mathbb{R}^3) \quad \text{and} \quad u(t) \in H^{3/2}(\mathbb{R}^3)$$

at each time in the closed interval. The critical quantities are then defined on the closed interval by

$$H(t) = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2, \quad E_{3/2}(t) = \frac{1}{2} \left\| \Lambda^{3/2} u(t) \right\|_2^2$$

The calculation of critical height led to these spatial levels; the intersection is what controls them at the endpoint.

Pressure is not an element of the velocity intersection. Write

$$Q_n = \partial_t^n p,$$

at every finite depth. The pressure Poisson equation recovers each Q_n from the velocity rows, while the differentiated momentum equation keeps ∇Q_n coupled to V_{n+1} , viscosity, and transport. Thus the pressure jet is recovered from u without treating pressure as an element of the velocity space.

Each estimate controls one finite block of this intersection. The two Sobolev levels needed to produce a common time trace determine the upper and lower edges of that block.

B.6 Finite Adjacent Rectangles

The whole intersection is reached through finite estimates, so begin with the first finite demand that cannot be met by one isolated Sobolev membership. Fix $I = (0, T)$ with $T < T_*$, and ask that both

$$V_0 = u \quad \text{and} \quad V_1 = \partial_t u$$

arrive at the same time slice in $H^2(\mathbb{R}^3)$. An $L_t^2 H_x^3$ bound for V_0 supplies spatial strength but no time trace by itself. The missing temporal side is

$$\partial_t V_0 = V_1 \in L^2(I; H^1).$$

To place V_1 at the same H^2 trace height, the same argument asks for

$$V_1 \in L^2(I; H^3), \quad \partial_t V_1 = V_2 \in L^2(I; H^1).$$

The four requirements are

$$V_0, V_1 \in L^2(I; H^3), \quad V_1, V_2 \in L^2(I; H^1).$$

The repeated V_1 is not redundant: it is the derivative of V_0 in the first trace pair and the field whose own derivative is V_2 in the second.

The trace mechanism explains why the two spatial levels must be adjacent around H^2 . Let

$$A := (1 - \Delta)^{1/2}, \quad \|v\|_{H^s} = \|A^s v\|_2.$$

For a smooth v ,

$$\|v(t)\|_{H^2}^2 - \|v(s)\|_{H^2}^2 = 2 \operatorname{Re} \int_s^t \left\langle A \partial_\tau v(\tau), A^3 v(\tau) \right\rangle_{L^2} d\tau.$$

The integrand satisfies

$$\left| \left\langle A \partial_\tau v, A^3 v \right\rangle_{L^2} \right| \leq \|\partial_\tau v\|_{H^1} \|v\|_{H^3},$$

which belongs to $L^1(I)$ by Cauchy–Schwarz when

$$v \in L^2(I; H^3), \quad \partial_t v \in L^2(I; H^1).$$

The standard density argument then yields

$$v \in C([0, T]; H^2(\mathbb{R}^3)).$$

Applied first to V_0 and then to V_1 , this gives

$$V_0, V_1 \in C([0, T]; H^2(\mathbb{R}^3)).$$

The two trace calculations use four norms, which we collect in

$$\begin{aligned} \mathfrak{R}_{1,2}(u; T) := & \|u\|_{L^2(0,T;H^3)}^2 + \|\partial_t u\|_{L^2(0,T;H^3)}^2 \\ & + \|\partial_t u\|_{L^2(0,T;H^1)}^2 + \|\partial_t^2 u\|_{L^2(0,T;H^1)}^2. \end{aligned}$$

Every entry is a derivative of the original solution. The rectangle groups the four norms needed for the two traces; it does not generate new velocity fields.

The same trace calculation determines the general cut. Fix independent integers $N, M \in \mathbb{N}_0$. The desired trace height is H^M , and its adjacent pair is

$$H^{M+1}(\mathbb{R}^3) \hookrightarrow H^M(\mathbb{R}^3) \hookrightarrow H^{M-1}(\mathbb{R}^3).$$

When $M = 0$, the lower member is the inhomogeneous negative space $H^{-1}(\mathbb{R}^3)$. For every $0 \leq n \leq N$, the trace of V_n at height H^M asks for

$$V_n \in L^2(0, T; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(0, T; H^{M-1}).$$

The squared velocity norm containing all of these adjacent pairs is

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned}$$

For a smooth representative, each retained temporal row obeys

$$\left| \frac{d}{dt} \|V_n(t)\|_{H^M}^2 \right| \leq 2 \|V_{n+1}(t)\|_{H^{M-1}} \|V_n(t)\|_{H^{M+1}}.$$

The right-hand side lies in $L^1(0, T)$, and density together with weak continuity yields

$$V_n \in C([0, T]; H^M(\mathbb{R}^3)), \quad 0 \leq n \leq N.$$

Increasing N retains more temporal responses. Increasing M raises the spatial trace height. The two choices remain independent, and the bound attached to the rectangle may depend on both.

The number $\mathfrak{R}_{N,M}(u; T)$ measures the velocity rows needed for those traces. To compare finite blocks without losing the equation that relates their entries, keep the corresponding pressure rows and differentiated equations as well. For the solution (u, p) generated by u_0 , define

$$\mathcal{R}_{N,M}[u_0; T] := \left((V_n)_{n=0}^{N+1}, (Q_n)_{n=0}^N; \mathfrak{R}_{N,M}(u; T) \right),$$

where

$$V_n = \partial_t^n u, \quad Q_n = \partial_t^n p.$$

For every $0 \leq n \leq N$, the retained rows satisfy

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n,$$

$$\nabla \cdot V_n = 0,$$

and

$$-\Delta Q_n = \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j.$$

Rapid decay fixes the normalization of every Q_n . At the retained Sobolev depths, each mixed velocity coordinate is the weak spatial derivative

$$J_{n,\alpha} = \partial_x^\alpha V_n,$$

and its pressure coordinate is the corresponding derivative of Q_n . Thus $\mathfrak{R}_{N,M}$ is the squared finite velocity norm, while $\mathcal{R}_{N,M}$ also retains the equations and normalized pressure rows that identify those velocities as derivatives of (u, p) . The same finite block can be restricted to different time intervals. For $J = (0, \tau)$ with $0 < \tau \leq T_*$, let $\mathcal{X}_{N,M}(J)$ denote the records whose velocity rows satisfy

$$W_n \in L^2(J; H^{M+1}), \quad W_{n+1} \in L^2(J; H^{M-1}) \quad (0 \leq n \leq N),$$

with $\partial_t W_n = W_{n+1}$ in distributions on $J \times \mathbb{R}^3$ for $0 \leq n \leq N$, whose normalized pressure rows satisfy

$$P_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (W_a)_i (W_{n-a})_j \in L^1(J; H^M) \quad (0 \leq n \leq N),$$

with $\nabla P_n \in L^1(J; H^{M-1})$, and whose retained rows obey the displayed momentum and divergence equations in distributions on $J \times \mathbb{R}^3$. The scalar component of such a record is the adjacent-row norm computed from its velocity rows on J . In particular,

$$\mathcal{R}_{N,M}[u_0; T] \in \mathcal{X}_{N,M}((0, T)).$$

As (N, M) ranges over the finite pairs, these rectangles cover every finite coordinate of the velocity intersection. Each rectangle has its own finite constant. Whenever two rectangles contain the same derivative, they must contain the same distribution; that agreement is what lets the finite blocks assemble.

C Compatible Terminal Assembly

Every whole-terminal rectangle contains derivatives of the same velocity and normalized pressure. The restriction maps make that agreement exact; the adjacent trace then produces one endpoint, the differentiated equation produces its complete jet, and local uniqueness joins the restarted solution to the history on the left.

C.1 The Terminal Quantifier

Keep the rapid-decay datum u_0 , the same viscosity $\nu > 0$, and the maximal classical velocity and pressure history (u, p) that they generate. Let its maximal lifespan be $[0, T_*)$, suppose for contradiction that $T_* < \infty$, and write $I_* = (0, T_*)$. Every derivative introduced below is a coordinate of this one history. The datum is smooth and divergence-free, so it belongs to every finite solenoidal Sobolev space. Write

$$H_\sigma^\infty(\mathbb{R}^3) := \left\{ v \in \bigcap_{m \in \mathbb{N}_0} H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0 \right\}. \quad (\text{C.1.1})$$

Because the history decays at spatial infinity, the whole-space normalization fixes pressure from the same velocity field. If $R_j = \partial_j(-\Delta)^{-1/2}$ denotes the j -th Riesz transform, then

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j). \quad (\text{C.1.2})$$

For $V_n = \partial_t^n u$ and $Q_n = \partial_t^n p$, direct differentiation of the original equation gives

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (\text{C.1.3})$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n. \quad (\text{C.1.4})$$

The binomial sum retains every way that the time derivatives divide across the transported and transporting velocity factors. At every temporal depth, transport, pressure, viscosity, and incompressibility remain joined inside the same evolution.

To inspect a finite part of the tower, choose independent temporal and spatial depths N and M . For finite $N, M \in \mathbb{N}_0$ and $0 < T < T_*$, define the squared adjacent-row norm

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|\partial_t^n u\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|\partial_t^{n+1} u\|_{L^2(0,T;H^{M-1})}^2. \end{aligned} \quad (\text{C.1.5})$$

Negative Sobolev indices are understood in the inhomogeneous Bessel-potential scale. The finite record $\mathcal{R}_{N,M}[u_0; T]$ consists of these velocity rows together with the pressure rows and differentiated equations that identify them as derivatives of the same history. Classical smoothness already gives every such record after a strict terminal time $T < T_*$ has been fixed.

Lemma C.1.1 (Rectangles on strict classical subintervals). *For every $T < T_*$ and every finite N, M , one has*

$$\mathfrak{R}_{N,M}(u; T) < \infty. \quad (\text{C.1.6})$$

If $N' \leq N$, $M' \leq M$, and $T' \leq T$, restriction of the larger rectangle gives the smaller rectangle.

Proof. The closed interval $[0, T]$ lies inside the classical lifespan. Smooth-data persistence gives $\partial_t^n u \in C([0, T]; H^q(\mathbb{R}^3))$ for every finite temporal order n and every finite Sobolev order q . Compactness of $[0, T]$ makes each of these norms bounded. In particular,

$$\|\partial_t^n u\|_{L^2(0, T; H^q)}^2 \leq T \sup_{0 \leq t \leq T} \|\partial_t^n u(t)\|_{H^q}^2 < \infty.$$

Choosing the spatial orders in (C.1.5) proves (C.1.6). Restriction preserves the solution and every one of its derivatives, so every shared coordinate agrees. $\square$

The gap between a strict cutoff and the whole terminal interval is already visible in one scalar function. If $f(t) = (T_* - t)^{-1/2}$, then

$$f \in L^2(0, T) \quad \text{for every } T < T_*, \quad f \notin L^2(I_*).$$

This function models only the quantifier and is not a Navier–Stokes history. It shows why smoothness on every strict subinterval cannot supply a whole-terminal bound.

The strict-subinterval lemma stops at $T < T_*$, and its bound may grow as $T \uparrow T_*$. The whole-terminal estimate in Theorem 3.2.1 supplies a bound independent of the cutoff after N and M have been fixed. To state the agreement between different finite depths, define, for $N' \leq N$ and $M' \leq M$, the terminal bonding map

$$\rho_{N', M'; I_*}^{N, M; I_*} : \mathcal{X}_{N, M}(I_*) \longrightarrow \mathcal{X}_{N', M'}(I_*)$$

by forgetting the higher rows, applying the Sobolev embeddings to every retained velocity and pressure row, and recomputing the scalar adjacent-row norm on I_* . Forgetting in two stages gives the same result as forgetting directly, and leaving the depths unchanged gives the identity map.

Proposition C.1.2 (Terminal realization of the rectangle estimate). *Assume the whole-terminal estimate in Theorem 3.2.1. For every finite $N, M \in \mathbb{N}_0$, the same datum-generated pressure-complete history satisfies*

$$\begin{aligned} \mathfrak{R}_{N,M}(u; I_*) &:= \sum_{n=0}^N \|\partial_t^n u\|_{L^2(I_*; H^{M+1})}^2 \\ &+ \sum_{n=0}^N \|\partial_t^{n+1} u\|_{L^2(I_*; H^{M-1})}^2 < \infty. \end{aligned} \quad (\text{C.1.7})$$

These finite coordinate bounds place the same mixed-jet record on the whole terminal interval:

$$\mathcal{R}_{N,M}[u_0] \in \mathcal{X}_{N,M}(I_*),$$

obtained from the same velocity and normalized pressure rows on I_ . If $N' \leq N$ and $M' \leq M$, the terminal restriction map forgets the higher temporal coordinates, applies the continuous Sobolev embeddings, and satisfies*

$$\rho_{N',M';I_*}^{N,M;I_*}(\mathcal{R}_{N,M}[u_0]) = \mathcal{R}_{N',M'}[u_0]. \quad (\text{C.1.8})$$

Every shared coordinate is the corresponding distributional derivative of the same u , and every pressure row uses the normalization in (C.1.2).

Proof. Fix N, M . For $0 \leq n \leq N$, set

$$f_n^+(t) := \|V_n(t)\|_{H^{M+1}}^2, \quad f_n^-(t) := \|V_{n+1}(t)\|_{H^{M-1}}^2.$$

These functions are nonnegative, and the intervals $(0, T)$ increase to I_* . For $\sigma \in \{+, -\}$,

$$\int_{I_*} f_n^\sigma(t) dt = \lim_{T \uparrow T_*} \int_0^T f_n^\sigma(t) dt.$$

Each integral on the right is a nonnegative term in the finite sum bounded by Theorem 3.2.1, so

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} \in L^2(I_*; H^{M-1}).$$

Adding these finitely many squared norms gives the velocity part of (C.1.7). Riesz-transform boundedness, the Sobolev product estimate, and Cauchy–Schwarz in time give, for $0 \leq n \leq N$,

$$\|Q_n\|_{L^1(I_*; H^M)} \leq C_{M,n} \sum_{a=0}^n \|V_a\|_{L^2(I_*; H^{M+1})} \|V_{n-a}\|_{L^2(I_*; H^{M+1})} < \infty.$$

This completes the pressure component of the record. Forgetting higher temporal rows or lowering the spatial index leaves every shared distributional derivative and normalized pressure row unchanged, which gives (C.1.8). $\square$

Take the rectangle $(N, M) = (2, 3)$. Its velocity coordinates are V_0, V_1, V_2 in $L^2(I_*; H^4)$ and V_1, V_2, V_3 in $L^2(I_*; H^2)$, together with the pressure rows Q_0, Q_1, Q_2 in $L^1(I_*; H^3)$. Restricting this rectangle to $(1, 2)$ drops V_3 and Q_2 , views V_0, V_1 in $L^2(I_*; H^3)$ and V_1, V_2 in $L^2(I_*; H^1)$, and

views Q_0, Q_1 in $L^1(I_*; H^2)$. Every retained coordinate is the same distributional derivative, now read in a weaker Sobolev space. This is compatibility; the scalar example above concerns the separate question of whether any one coordinate stays finite on the whole terminal interval.

The finite-rectangle form also recovers the adjacent rows coordinate by coordinate.

Proposition C.1.3 (Adjacent-row recovery). *The whole-terminal estimate gives, for every finite $n, m \in \mathbb{N}_0$,*

$$V_n \in L^2(I_*; H^{m+1}), \quad \partial_t V_n = V_{n+1} \in L^2(I_*; H^{m-1}). \quad (\text{C.1.9})$$

Each finite pair has its own constant.

Proof. Fix n and m , and choose the rectangle with $N = n$ and $M = m$. The two terms indexed by n in the whole-terminal sum (C.1.7) are finite, which gives the memberships in (C.1.9). Every row in (C.1.9) is an actual derivative of u , and (C.1.8) preserves that identity under restriction. $\square$

The strict and whole-terminal rectangles use the same distributional derivatives of (u, p) . Equation (C.1.8) makes the full family compatible under every decrease of N or M .

C.2 The Compatible Intersection

For each finite pair (N, M) , the whole-terminal estimate bounds the corresponding mixed-jet rectangle on I_* . If N or M is increased, the larger rectangle contains every derivative in the smaller one, and each shared entry is the same distributional derivative of u . Order the finite pairs coordinatewise and use the bonding maps from (C.1.8). The compatible families form

$$\mathcal{X}_\infty(I_*) := \varprojlim_{N, M} \mathcal{X}_{N, M}(I_*).$$

A family $(\mathbf{R}_{N, M})_{N, M}$ belongs to this set exactly when $\mathbf{R}_{N, M} \in \mathcal{X}_{N, M}(I_*)$ for every finite pair and

$$\rho_{N', M'; I_*}^{N, M; I_*}(\mathbf{R}_{N, M}) = \mathbf{R}_{N', M'}$$

whenever $N' \leq N$ and $M' \leq M$.

Proposition C.2.1 (Compatible projective assembly). *The terminal rectangles determine the datum-specific inverse-limit element*

$$\mathcal{I}[u_0] := (\mathcal{R}_{N, M}[u_0])_{N, M}.$$

This family belongs to $\mathcal{X}_\infty(I_)$. Its velocity realization is the same distribution u in every coordinate and satisfies*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (\text{C.2.1})$$

Every coordinate projection commutes with every restriction map.

Proof. Fix finite $r, m \in \mathbb{N}_0$. Choose $N \geq r$ and $M \geq m$. The rectangle $\mathcal{R}_{N,M}[u_0]$ contains

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

These are the distributional time derivatives of the same velocity field, so $u \in H^r(I_*; H^m)$. The arbitrariness of r and m gives (C.2.1). Equation (C.1.8) makes every lower record the projection of every larger record. The compatible family thus determines one inverse-limit element with no representative choices to reconcile. $\square$

Taking the zeroth temporal projection means retaining u itself while allowing the spatial depth to range through every finite order. Define

$$\mathcal{S}_0(I_*) := \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (\text{C.2.2})$$

For each m , the projection selects the velocity coordinate of $\mathcal{R}_{0,m}[u_0]$. Compatibility says that all of these coordinates are the same u , viewed in different Sobolev spaces. Consequently,

$$\pi_0(\mathcal{I}[u_0]) = u \in \mathcal{S}_0(I_*).$$

This projection also contains every finite spatial energy read by the tower. For $\Lambda = (-\Delta)^{1/2}$ and $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2. \quad (\text{C.2.3})$$

Choosing an integer $m \geq s$ in (C.2.2) gives

$$\int_0^{T_*} E_s(t) dt \leq C_{s,m} \|u\|_{L^2(I_*; H^m)}^2 < \infty. \quad (\text{C.2.4})$$

The intersection keeps these finite Sobolev norms as separate coordinates of the same u .

C.3 The Critical Height at the Terminal Face

The whole-terminal rectangles contain adjacent spatial rows on I_* . Before using them at critical height, we prove that one such pair gives a strong value at T_* .

Lemma C.3.1 (Adjacent Hilbert-scale trace). *Let $s \in \mathbb{R}$, and let v satisfy*

$$v \in L^2(I_*; H^{s+1}), \quad v_t \in L^2(I_*; H^{s-1}).$$

Then v has a unique representative in $C([0, T_]; H^s)$. For $0 \leq a \leq b \leq T_*$, this representative satisfies*

$$\left| \|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 \right| \leq 2 \|v_t\|_{L^2(a,b; H^{s-1})} \|v\|_{L^2(a,b; H^{s+1})}. \quad (\text{C.3.1})$$

Proof. Put $A = (1 - \Delta)^{1/2}$. Time mollification and Fourier truncation first give, for smooth approximants,

$$\|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 = 2 \operatorname{Re} \int_a^b \langle A^{s-1} v_t(t), A^{s+1} v(t) \rangle_{L^2} dt. \quad (\text{C.3.2})$$

Cauchy's inequality gives (C.3.1) for the approximants. The two assumed L^2 norms let the approximants converge in the right side of (C.3.2). The resulting identity makes the H^s norm continuous, while the same approximation gives weak H^s continuity. Weak continuity together with norm continuity gives strong H^s continuity. Two continuous representatives that agree almost everywhere agree everywhere, which proves uniqueness. $\square$

The critical readout uses only the rectangle $\mathcal{R}_{0,2}[u_0]$. That rectangle supplies $u \in L^2(I_*; H^3)$ and $u_t \in L^2(I_*; H^1)$, hence

$$u \in L^2(I_*; H^{5/2}), \quad u_t \in L^2(I_*; H^{1/2}). \quad (\text{C.3.3})$$

The fractional pair follows from the continuous inhomogeneous Sobolev embeddings, so no fractional-index rectangle is being assumed. Applying the adjacent trace lemma with $s = 3/2$ gives

$$u \in C([0, T_*]; H^{3/2}). \quad (\text{C.3.4})$$

Let $H(t) = E_{1/2}(t)$, with E_s defined in (C.2.3). The homogeneous identity needed here follows directly on the critical pair. Apply time mollification together with Fourier multipliers that truncate to $\varepsilon \leq |\xi| \leq R$. For the resulting smooth, frequency-localized functions, differentiation and Plancherel's theorem give

$$\frac{1}{2} \left\| \Lambda^{3/2} u(b) \right\|_{L^2}^2 - \frac{1}{2} \left\| \Lambda^{3/2} u(a) \right\|_{L^2}^2 = \operatorname{Re} \int_a^b \left\langle \Lambda^{1/2} u_t(t), \Lambda^{5/2} u(t) \right\rangle_{L^2} dt.$$

The two memberships in (C.3.3) give convergence of the integral by Cauchy's inequality. The continuity in (C.3.4) gives convergence of the endpoint terms as $\varepsilon \downarrow 0$, $R \uparrow \infty$, and the time mollifier is removed. Consequently, for $0 \leq a \leq b \leq T_*$,

$$E_{3/2}(b) - E_{3/2}(a) = \operatorname{Re} \int_a^b \left\langle \Lambda^{1/2} u_t(t), \Lambda^{5/2} u(t) \right\rangle_{L^2} dt.$$

The pair in (C.3.3) places the integrand in $L^1(I_*)$. Equation (C.2.4) supplies $E_{3/2} \in L^1(I_*)$. Thus

$$E_{3/2} \in W^{1,1}(I_*) \cap L^1(I_*) \subset L^\infty(I_*). \quad (\text{C.3.5})$$

The stronger integer rows $u, u_t \in L^2(I_*; H^1)$ give

$$H'(t) = \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle_{L^2} \in L^1(I_*). \quad (\text{C.3.6})$$

The pressure-explicit critical production current is

$$\mathcal{P}_{1/2} := -\operatorname{Re} \left\langle \Lambda^{1/2} u, \Lambda^{1/2} \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

Pairing the original equation with Λu yields the exact balance

$$H' = -2\nu E_{3/2} + \mathcal{P}_{1/2}.$$

Equations (C.3.5) and (C.3.6) now give

$$H \in W^{1,1}(I_*), \quad E_{3/2} \in L^\infty(I_*) \cap L^1(I_*), \quad \mathcal{P}_{1/2} \in L^1(I_*).$$

Thus the critical height and $E_{3/2}$ have continuous representatives up to T_* , while the pressure-explicit production current is integrable in time. No endpoint value is assigned to that current. The rectangle $\mathcal{R}_{0,2}[u_0]$ gives this critical trace. The higher rectangles place the same u in every spatial Sobolev row, which is the input needed to obtain one terminal value in every H^m .

C.4 One Endpoint for the Whole Tower

A smooth restart requires one value $u(T_*)$ in every finite Sobolev space. The complete spatial row gives the required space integrability, and the original equation converts it into enough time integrability to reach T_* .

Lemma C.4.1 (The complete spatial row reaches every endpoint topology). *Let (u, p) be the fixed pressure-normalized Navier–Stokes history on I_* . If $u \in \mathcal{S}_0(I_*)$, then, for every $m \in \mathbb{N}_0$,*

$$u_t \in L^1(I_*; H^m), \quad u \in W^{1,1}(I_*; H^m) \hookrightarrow C([0, T_*]; H^m). \quad (\text{C.4.1})$$

Proof. The Leray projection rewrites the pressure-completed interaction as

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u).$$

Under the fixed normalization, the Leray expression is exactly the combined transport-plus-pressure response. Sobolev multiplication in three dimensions and the boundedness of the Leray and Riesz multipliers give

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{H^m} \leq C_m \|u\|_{H^{m+2}}^2.$$

The complete spatial row and the finiteness of T_* yield

$$\begin{aligned} \|\mathbb{P} \nabla \cdot (u \otimes u)\|_{L^1(I_*; H^m)} &\leq C_m \|u\|_{L^2(I_*; H^{m+2})}^2 < \infty, \\ \|\Delta u\|_{L^1(I_*; H^m)} &\leq T_*^{1/2} \|u\|_{L^2(I_*; H^{m+2})} < \infty. \end{aligned}$$

The momentum equation then places u_t in $L^1(I_*; H^m)$. More concretely, for $0 \leq s < t < T_*$,

$$\|u(t) - u(s)\|_{H^m} \leq \nu(t-s)^{1/2} \|u\|_{L^2((s,t); H^{m+2})} + C_m \|u\|_{L^2((s,t); H^{m+2})}^2.$$

Both terms tend to zero as $s, t \uparrow T_*$, so the same velocity is Cauchy in H^m and reaches one endpoint there. The inclusion $L^2(I_*; H^m) \subset L^1(I_*; H^m)$ places u in the Bochner space $W^{1,1}(I_*; H^m)$. The Bochner fundamental theorem of calculus supplies the continuous representative in (C.4.1). $\square$

For each m , let $u_*^{(m)} \in H^m$ denote the endpoint value supplied by (C.4.1). If $k > m$, the H^k -continuous representative, viewed in H^m , and the H^m -continuous representative agree almost everywhere on I_* . Their continuity makes them identical on the closed interval. Their terminal values are consequently compatible and define one element

$$u_* := u(T_*) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (\text{C.4.2})$$

No estimate uniform in m occurs in this conclusion. Continuity of divergence from H^m to H^{m-1} makes u_* divergence-free. The strong L^2 trace gives u_* finite kinetic energy.

Apply the adjacent trace lemma to each temporal row in (C.1.9). It gives

$$\partial_t^N u \in C([0, T_*]; H^m) \quad (N, m \in \mathbb{N}_0).$$

For every $N \in \mathbb{N}_0$, compatibility across the spatial orders identifies these traces as one value

$$V_N^* := \lim_{t \uparrow T_*} \partial_t^N u(t) \in H^\infty(\mathbb{R}^3). \quad (\text{C.4.3})$$

The differentiated equation now identifies these traces with the jet generated from u_* .

Proposition C.4.2 (Pressure-complete endpoint jet). *The endpoint velocity u_* determines a unique sequence $(V_N^*, Q_N^*)_{N \geq 0} \subset H^\infty \times H^\infty$ through*

$$\begin{aligned} V_0^* &= u_*, \\ Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \end{aligned} \quad (\text{C.4.4})$$

$$V_{N+1}^* = \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \quad (\text{C.4.5})$$

This recursively generated sequence equals the left endpoint traces of the original velocity and pressure history.

Proof. The Fréchet space $H^\infty = \bigcap_m H^m$ is closed under spatial differentiation and multiplication. Riesz transforms preserve every H^m , so they preserve H^∞ . Equations (C.4.4) and (C.4.5) thus construct one unique pair at each finite order.

Fix a target Sobolev order m . The traces in (C.4.3) converge in Sobolev spaces high enough for every product and derivative appearing in the recurrence. Multiplication, spatial differentiation, and the Riesz transforms are continuous between those finite Sobolev spaces. Passing $t \uparrow T_*$ in (C.1.3) and (C.1.4) gives (C.4.4) and (C.4.5). Induction in N , beginning with $V_0^* = u_*$, identifies every recursively generated coordinate with the corresponding trace of the history. The arbitrariness of m gives the identities in H^∞ . Spatial differentiation then identifies every mixed coordinate $\partial_t^N \partial_x^\alpha u$ at the terminal face. $\square$

The pressure limit has no new additive constant to choose at T_* . The decaying Riesz formula fixes $Q_N(t)$ for every $t < T_*$, and passage to the limit preserves that normalization. Hence every left limit $\partial_t^N \partial_x^\alpha u(T_*)$ is exactly the derivative obtained by inserting u_* into the pressure-normalized Navier–Stokes recurrences.

C.5 Restarting the Same History

Set the restart datum equal to the terminal value u_* . Local well-posedness gives a solution on the right of T_* , and persistence keeps every finite Sobolev order on one common restart interval. Both one-sided jets are generated from u_* by the same normalized recurrences, so they agree at the joining time; uniqueness then identifies the two pieces as one solution.

Lemma C.5.1 (Smooth local restart and uniqueness). *Let $t_0 \geq 0$ and let $a \in H_\sigma^\infty(\mathbb{R}^3)$, with the class defined in (C.1.1). There is a $\delta > 0$ and a unique smooth pressure-normalized solution (w, q) of the same Navier–Stokes system on $[t_0, t_0 + \delta] \times \mathbb{R}^3$ with $w(t_0) = a$ and*

$$q = R_i R_j (w_i w_j).$$

Every finite spatial Sobolev order persists on the same interval, and the temporal jet at t_0 is generated by the pressure-normalized recurrences (C.4.4) and (C.4.5).

Proof. By time translation, Proposition A.1.1 applies at t_0 with datum a . Its H^3 -based lifespan depends only on ν and $\|a\|_{H^3}$; since $a \in H_\sigma^\infty$, its persistence clause places every finite Sobolev order on that same interval and supplies the unique normalized-pressure solution. Differentiating the smooth solution at its initial face and using (C.1.2) gives the stated pressure and velocity recurrences. $\square$

Theorem C.5.2 (Global continuation from whole-terminal rectangles). *Let u_0 be smooth and divergence-free on $\mathbb{R}^3$, with u_0 and all of its spatial derivatives decaying rapidly at infinity, let $\nu > 0$, and let (u, p) be the corresponding maximal pressure-normalized classical history. Assume that this history satisfies the whole-terminal compatible-rectangle conclusion of Proposition C.1.2. Then $T_* = \infty$, and its normalized velocity and pressure are smooth on every finite time interval.*

Proof. Assume for contradiction that the maximal time T_* is finite. By the theorem’s hypothesis, every finite record $\mathcal{R}_{N,M}[u_0]$ exists on I_* , with compatible restrictions. Their compatible intersection gives the common endpoint $u_* \in H_\sigma^\infty(\mathbb{R}^3)$ in (C.4.2) and the actual pressure-complete endpoint jet in (C.4.4)–(C.4.5). Apply the local restart lemma at $t_0 = T_*$ with datum $a = u_*$. It produces a smooth solution (w, q) on $[T_*, T_* + \delta]$. Define

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w, q)(t), & T_* \leq t \leq T_* + \delta. \end{cases}$$

The velocity values agree at T_* by construction. The left and right pressure and velocity derivatives agree at every finite order because both jets obey the same normalized recurrences from the same endpoint datum. The piecewise pair is consequently smooth across T_* and solves the same fixed-viscosity system there. Its lifespan is longer than the alleged maximal lifespan. Strong-solution uniqueness identifies this extension as the continuation of the original datum-generated history. This contradicts the definition of T_* . Hence

$$T_* = \infty.$$

Every finite temporal and spatial derivative remains in the compatible Sobolev intersection on each finite interval, and Sobolev embedding gives

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).$$

$\square$

Bibliography

- Bahouri, Hajer, Jean-Yves Chemin, and Raphaël Danchin (2011). *Fourier Analysis and Nonlinear Partial Differential Equations*. Vol. 343. Grundlehren der mathematischen Wissenschaften. Berlin and Heidelberg: Springer. DOI: 10.1007/978-3-642-16830-7.
- Beale, J. Thomas, Tosio Kato, and Andrew Majda (1984). “Remarks on the Breakdown of Smooth Solutions for the 3-D Euler Equations”. In: *Communications in Mathematical Physics* 94.1, pp. 61–66. DOI: 10.1007/BF01212349.
- Beirão da Veiga, Hugo (1995). “A New Regularity Class for the Navier–Stokes Equations in $\mathbb{R}^n$ ”. In: *Chinese Annals of Mathematics, Series B* 16.4, pp. 407–412.
- Caffarelli, Luis, Robert V. Kohn, and Louis Nirenberg (1982). “Partial regularity of suitable weak solutions of the Navier–Stokes equations”. In: *Communications on Pure and Applied Mathematics* 35.6, pp. 771–831. DOI: 10.1002/cpa.3160350604.
- Cerdà, Joan (2010). *Linear Functional Analysis*. Vol. 116. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society. ISBN: 978-0-8218-5115-9.
- Escauriaza, L., G. A. Seregin, and V. Šverák (2003). “ $L_{3,\infty}$ -solutions of the Navier–Stokes equations and backward uniqueness”. In: *Russian Mathematical Surveys* 58.2, pp. 211–250. DOI: 10.1070/RM2003v058n02ABEH000609.
- Fefferman, Charles L. (2000). *Existence and Smoothness of the Navier–Stokes Equation*. Clay Mathematics Institute. URL: <https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf> (visited on 08/13/2026).
- Fujita, Hiroshi and Tosio Kato (1964). “On the Navier–Stokes Initial Value Problem. I”. In: *Archive for Rational Mechanics and Analysis* 16.4, pp. 269–315. DOI: 10.1007/BF00276188.
- Kato, Tosio (1984). “Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions”. In: *Mathematische Zeitschrift* 187.4, pp. 471–480. DOI: 10.1007/BF01174182.
- Ladyzhenskaya, O. A. (1967). “On the uniqueness and on the smoothness of weak solutions of the Navier–Stokes equations”. In: *Zapiski Nauchnykh Seminarov LOMI* 5, pp. 169–185. URL: <https://www.mathnet.ru/eng/zns12228>.
- Leray, Jean (1934). “Sur le mouvement d’un liquide visqueux emplissant l’espace”. In: *Acta Mathematica* 63, pp. 193–248. DOI: 10.1007/BF02547354.
- Prodi, Giovanni (1959). “Un teorema di unicità per le equazioni di Navier–Stokes”. In: *Annali di Matematica Pura ed Applicata* 48.1, pp. 173–182. DOI: 10.1007/BF02410664.
- Seregin, Gregory (2012). “A Certain Necessary Condition of Potential Blow Up for Navier–Stokes Equations”. In: *Communications in Mathematical Physics* 312.3, pp. 833–845. DOI: 10.1007/s00220-011-1391-x.
- Serrin, James (1962). “On the interior regularity of weak solutions of the Navier–Stokes equations”. In: *Archive for Rational Mechanics and Analysis* 9, pp. 187–195. DOI: 10.1007/BF00253344.
- (1963). “The Initial Value Problem for the Navier–Stokes Equations”. In: *Nonlinear Problems*. Madison, WI: University of Wisconsin Press, pp. 69–98.
- Tao, Terence (2021). “Quantitative Bounds for Critically Bounded Solutions to the Navier–Stokes Equations”. In: *Nine Mathematical Challenges—An Elucidation*. Vol. 104. Proceedings of Symposia in Pure Mathematics. Providence, RI: American Mathematical Society, pp. 149–193. DOI: 10.1090/pspum/104/01874.